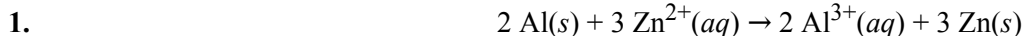
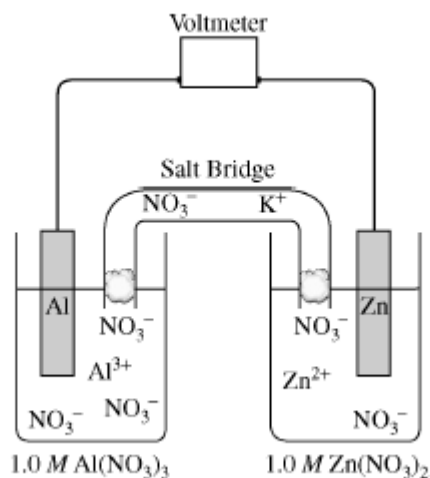


Chapter 12 Part 1: Galvanic Cell



- Write the complete electron configuration (e.g., $1s^2 2s^2 \dots$) for Zn^{2+} .
- Which species, Zn or Zn^{2+} , has the greater ionization energy? Justify your answer.
- Identify the species that is oxidized in the reaction.

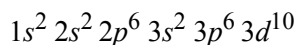
The diagram below shows a galvanic cell based on the reaction. Assume that the temperature is 25°C .



- The diagram includes a salt bridge that is filled with a saturated solution of KNO_3 . Describe what happens in the salt bridge as the cell operates.
- Determine the value of the standard voltage, E° , for the cell.
- Indicate whether the value of the standard free-energy change, ΔG° , for the cell reaction is positive, negative, or zero. Justify your answer.
- If the concentration of $\text{Al}(\text{NO}_3)_3$ in the $\text{Al}(s)/\text{Al}^{3+}(aq)$ half-cell is lowered from 1.0 M to 0.01 M at 25°C , does the cell voltage increase, decrease, or remain the same? Justify your answer.

Part A

1 point is earned for the correct configuration.



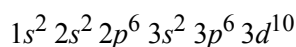
Chapter 12 Part 1: Galvanic Cell



0	1
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The student response earns one of the following points:

1 point is earned for the correct configuration.

**Part B**

1 point is earned for identifying Zn^{2+} with justification.

Zn^{2+} has the greater ionization energy. The electron being removed from Zn^{2+} experiences a larger effective nuclear charge than the electron being removed from Zn because Zn^{2+} has two fewer electrons shielding the nucleus.

OR

It takes more energy to remove a negatively charged electron from a positive ion than from a neutral atom.



0	1
---	---

The student response earns one of the following points:

1 point is earned for identifying Zn^{2+} with justification.

Zn^{2+} has the greater ionization energy. The electron being removed from Zn^{2+} experiences a larger effective nuclear charge than the electron being removed from Zn because Zn^{2+} has two fewer electrons shielding the nucleus.

OR

It takes more energy to remove a negatively charged electron from a positive ion than from a neutral atom.

Part C

1 point is earned for identifying Al.

Al(s)



0	1
---	---

The student response earns one of the following points:

Chapter 12 Part 1: Galvanic Cell

1 point is earned for identifying Al.

Al(s)

Part D

1 point is earned for correctly indicating the direction of ion flow.

As the cell operates, NO_3^- ions flow toward the Al half-cell and K^+ ions flow toward the Zn half-cell.



0	1
---	---

The student response earns one of the following points:

1 point is earned for correctly indicating the direction of ion flow.

As the cell operates, NO_3^- ions flow toward the Al half-cell and K^+ ions flow toward the Zn half-cell.

Part E

1 point is earned for the correct E° .

$$E^\circ = (-0.76 \text{ V}) - (-1.66 \text{ V}) = 0.90 \text{ V}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct E° .

$$E^\circ = (-0.76 \text{ V}) - (-1.66 \text{ V}) = 0.90 \text{ V}$$

Part F

1 point is earned for indicating that ΔG° is negative.

1 point is earned for a correct justification.

ΔG° is negative since E° is positive and $\Delta G^\circ = -nFE^\circ$

OR

ΔG° must be negative because the reaction is spontaneous under standard conditions.

Chapter 12 Part 1: Galvanic Cell



0	1	2
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The student response earns two of the following points:

1 point is earned for indicating that ΔG° is negative.

1 point is earned for a correct justification.

ΔG° is negative since E° is positive and $\Delta G^\circ = -nFE^\circ$

OR

ΔG° must be negative because the reaction is spontaneous under standard conditions.

Part G

1 point is earned for indicating that E_{cell} increases.

1 point is earned for the correct justification.

Lowering $[Al^{3+}]$ causes an increase in the cell voltage. The value of Q will fall below 1.0 and the log term in the Nernst equation will become negative. This causes the value of E_{cell} to become more positive.

OR

A decrease in a product concentration will increase the spontaneity of the reaction, increasing the value of E_{cell} .



0	1	2
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The student response earns two of the following points:

1 point is earned for indicating that E_{cell} increases.

1 point is earned for the correct justification.

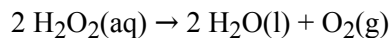
Lowering $[Al^{3+}]$ causes an increase in the cell voltage. The value of Q will fall below 1.0 and the log term in the Nernst equation will become negative. This causes the value of E_{cell} to become more positive.

OR

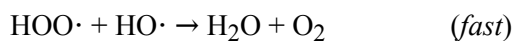
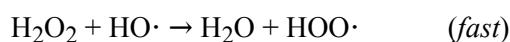
A decrease in a product concentration will increase the spontaneity of the reaction, increasing the value of E_{cell} .

Chapter 12 Part 1: Galvanic Cell

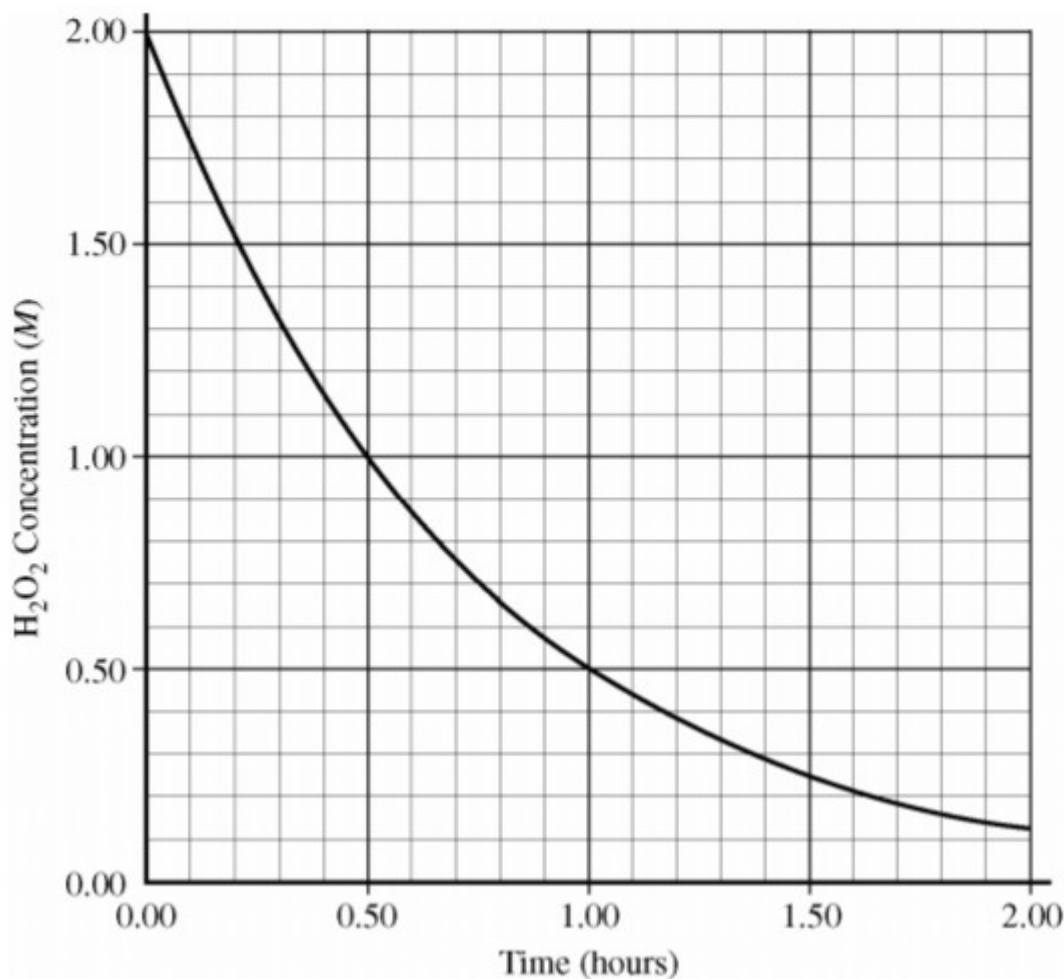
2.



The decomposition of hydrogen peroxide to form water and oxygen gas is represented by the equation above. A proposed mechanism for the reaction, which involves the free radicals $\text{HO}\cdot$ and $\text{HOO}\cdot$, is represented by the three equations below.



- Write the rate law consistent with the proposed mechanism above.
- The rate of the decomposition reaction was studied in an experiment, and the resulting data were plotted in the graph below.



Using the graph, determine the time, in hours, needed for the concentration of H_2O_2 to change from

- 1.50 M to 0.75 M

Chapter 12 Part 1: Galvanic Cell

- ii. 0.80 *M* to 0.40 *M*
- c. The experimental data are consistent with the proposed mechanism. Explain.

An electrochemical cell based on the decomposition of H_2O_2 can be constructed based on the half-reactions in the table below.

Half-Reaction	Standard Reduction Potential, E°
$\text{H}_2\text{O}_2 + 2 e^- \rightarrow 2 \text{OH}^-$	0.88 V
$\text{O}_2 + 2 \text{H}_2\text{O} + 2 e^- \rightarrow \text{H}_2\text{O}_2 + 2 \text{OH}^-$	-0.15 V

- d. Calculate the value of the standard cell potential, E° , for the cell.
- e. Indicate whether ΔG° for the decomposition reaction is greater than 0, less than 0, or equal to 0. Justify your answer.
- f. The decomposition of $\text{H}_2\text{O}_2(aq)$ is slow at 298 K, but a suitable catalyst greatly increases the rate of the decomposition reaction.
- i. Draw a circle around each of the quantities below that has a different value for the catalyzed reaction than for the uncatalyzed reaction.
- K_{eq} ΔG° ΔH° E_a
- ii. For any quantity that you circled above, indicate whether its value is greater or less for the catalyzed reaction than for the uncatalyzed reaction. Explain why.

Part A

1 point is earned for the correct rate law.

$$\text{rate} = k [\text{H}_2\text{O}_2]$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct rate law.

Chapter 12 Part 1: Galvanic Cell

$$\text{rate} = k [\text{H}_2\text{O}_2]$$

Part B

- 1 point is earned for the correct time.

$$\text{Time} = 0.71 - 0.21 = 0.50 \pm 0.05 \text{ hour}$$

- 1 point is earned for the correct time.

$$\text{Time} = 1.17 - 0.66 = 0.51 \pm 0.05 \text{ hour}$$



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for the correct time.

$$\text{Time} = 0.71 - 0.21 = 0.50 \pm 0.05 \text{ hour}$$

- 1 point is earned for the correct time.

$$\text{Time} = 1.17 - 0.66 = 0.51 \pm 0.05 \text{ hour}$$

Part C

1 point is earned for a correct explanation.

The data show that the time remains constant when the concentration of the reactant decreases by one-half, indicating that the reaction is first order.

The rate law for the proposed mechanism has $[\text{H}_2\text{O}_2]$ to the first power, indicating a first-order reaction.



0	1
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The student response earns one of the following points:

1 point is earned for a correct explanation.

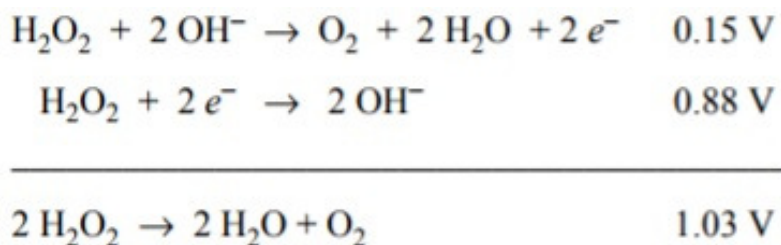
The data show that the time remains constant when the concentration of the reactant decreases by one-half, indicating that the reaction is first order.

The rate law for the proposed mechanism has $[\text{H}_2\text{O}_2]$ to the first power, indicating a first-order reaction.

Part D

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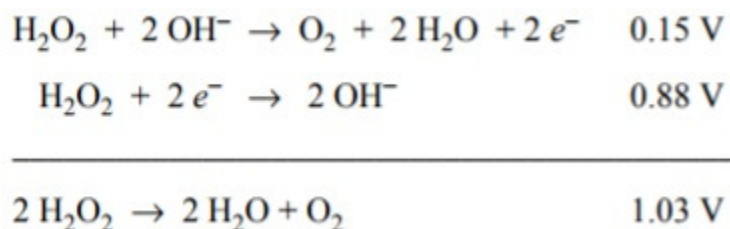
1 point is earned for the correct value of E° .



0	1
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The student response earns one of the following points:

1 point is earned for the correct value of E° .



Part E

1 point is earned for the correct answer with justification.

ΔG° is less than 0.

$\Delta G^\circ = -nFE^\circ$. Because E° is positive, ΔG° must be negative.

OR

E° is positive, thus the reaction is spontaneous. For spontaneous reactions, ΔG° is always negative.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

ΔG° is less than 0.

$\Delta G^\circ = -nFE^\circ$. Because E° is positive, ΔG° must be negative.

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OR

E° is positive, thus the reaction is spontaneous. For spontaneous reactions, ΔG° is always negative.

Part F

- 1 point is earned for circling E_a and none of the other quantities.

Only E_a should be circled.

- 1 point is earned for a correct answer with reference to an altered mechanism.

E_a , the activation energy, is less for the catalyzed reaction than for the uncatalyzed reaction. A catalyst provides a new reaction mechanism that requires a lower minimum required potential energy for the reaction to occur. Thus the activation energy is lowered. Note: reference to other circled quantities is ignored.



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for circling E_a and none of the other quantities.

Only E_a should be circled.

- 1 point is earned for a correct answer with reference to an altered mechanism.

E_a , the activation energy, is less for the catalyzed reaction than for the uncatalyzed reaction. A catalyst provides a new reaction mechanism that requires a lower minimum required potential energy for the reaction to occur. Thus the activation energy is lowered. Note: reference to other circled quantities is ignored.

3.	Half-Reaction	Standard Reduction Potential (V)
	$\text{Al}(\text{OH})_3(s) + 3 e^- \rightarrow \text{Al}(s) + 3 \text{OH}^-(aq)$	-2.31
	$\text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4 e^- \rightarrow 4 \text{OH}^-(aq)$	+0.40

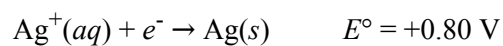
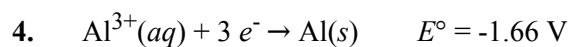
The table above provides the half-reactions and standard reduction potentials needed to determine if an Al-air fuel cell can be constructed. Which of the following equations best supports the possibility of obtaining electrical energy from such a cell?

(A) $E^\circ_{\text{cell}} = [+0.40 - (-2.31)] \text{ V}$ and $\Delta G^\circ_{\text{rxn}} = -\frac{(12 \times 96,500 \times 2.71)}{1,000} \text{ kJ}$ ✓

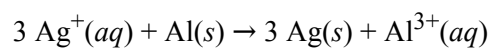
(B) $E^\circ_{\text{cell}} = [+0.40 - (-2.31)] \text{ V}$ and $\Delta G^\circ_{\text{rxn}} = -\frac{(4 \times 96,500 \times 2.71)}{1,000} \text{ kJ}$

(C) $E^\circ_{\text{cell}} = [-2.31 - (+0.40)] \text{ V}$ and $\Delta G^\circ_{\text{rxn}} = -\frac{(12 \times 96,500 \times -2.71)}{1,000} \text{ kJ}$

(D) $E^\circ_{\text{cell}} = [-2.31 - (+0.40)] \text{ V}$ and $\Delta G^\circ_{\text{rxn}} = -\frac{3 \times 96,500 \times -2.71}{1,000} \text{ kJ}$

Chapter 12 Part 1: Galvanic Cell

According to the standard reduction potentials given above, what is the standard cell potential for the reaction represented below?



(A) -1.74 V

(B) -0.86 V

(C) +1.74 V

(D) +2.46 V

(E) +4.06 V



Chapter 12 Part 1: Galvanic Cell

5. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions relating to the element aluminum, Al.

- (a) Write the complete ground-state electron configuration of an Al atom.
- (b) Based on principles of atomic structure, explain why the radius of the Al atom is larger than the radius of the Al^{3+} ion.

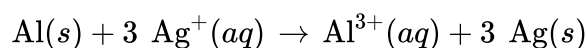
A student plans to combine solid aluminum with an aqueous solution of silver ions. The student determines the mass of solid AgNO_3 needed to prepare the solution with a specific concentration.

- (c) In the following table, briefly list the steps necessary to prepare 200.0 mL of an aqueous solution of AgNO_3 using only equipment selected from the choices given. Assume that all appropriate safety measures are already in place. Not all equipment or lines in the table may be needed.

- | | | |
|-------------------------|------------------------------|------------------|
| • Solid AgNO_3 | • Weighing paper and scoop | • 250 mL beakers |
| • Distilled water | • 200.00 mL volumetric flask | • Pipet |
| • Balance | • 50.0 mL graduated cylinder | |

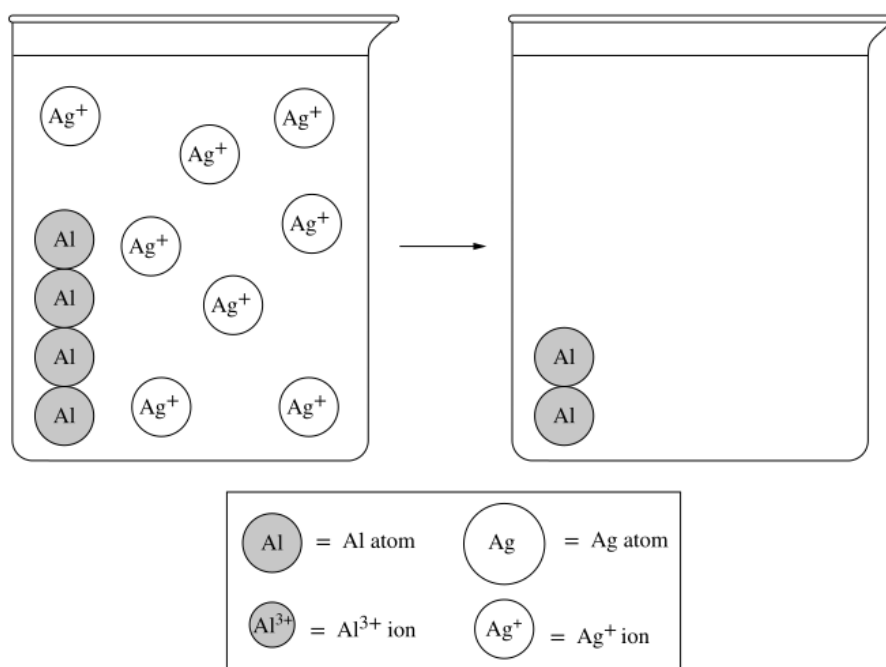
Step	Step Description
1.	Use weighing paper to measure the determined mass of solid AgNO_3 on a balance.
2.	

After preparing the solution, the student places some of the solution into a beaker and adds a sample of aluminum. The reaction represented by the following equation occurs.



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(d) The following diagram gives an incomplete particulate representation of the reaction. The beaker on the left represents the system before the mixture reacts. Complete the drawing on the right to represent the system after the reaction has occurred. Be sure to include 1) the correct type and number of particles based on the number shown on the left and 2) the relative spacing to depict the appropriate phases.



The student finds the standard reduction potentials given in the table, which are related to the reaction that occurs.

Half-Reaction	E°
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	0.80V
$\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$	-1.66V

- (e) Using the standard reduction potentials, calculate the value of E° for the reaction.
- (f) Based on the value of E° , would the standard free energy change of the reaction under standard conditions, ΔG° , be positive, negative, or zero? Justify your answer.
- (g) Once the reaction appears to stop progressing, would the change in free energy, ΔG , be positive, negative, or zero? Justify your answer.

Part A

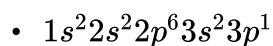
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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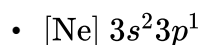


0	1
---	---

The response provides a correct electron configuration.



OR



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

- The highest occupied electron shell ($n = 3$) of Al is at a greater average distance from the nucleus than the highest occupied electron shell ($n = 2$) of Al^{3+} .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **both** of the following criteria.

- ☐ The response lists the correct steps to dissolve the solute in water (steps may be consolidated).
 - 2. Partially fill the volumetric flask with some distilled water
 - 3. Add the weighed $\text{AgNO}_3(s)$ to the volumetric flask
 - 4. Swirl to dissolve the solid
- ☐ The response provides the correct step to ensure quantitative dilution.

Chapter 12 Part 1: Galvanic Cell

- 5. After the solid is dissolved, fill the flask to the calibration (200.00 mL) mark and mix.

Part D

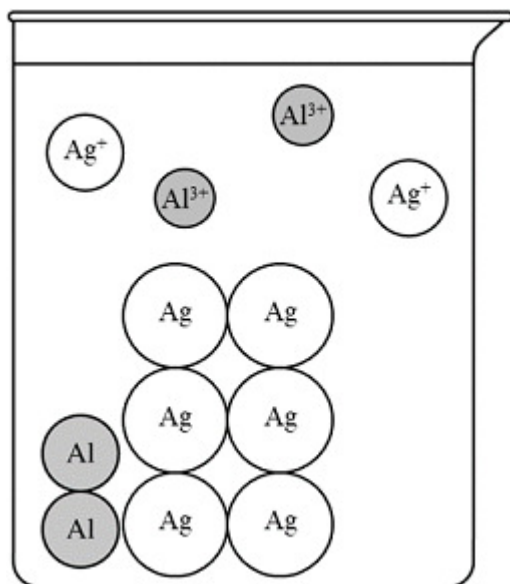
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The response meets **all** of the following criteria.

- ☐ The drawing shows product formation and indicates the conservation of matter.
 - 4 Al and 8 Ag particles in the beaker on right (see sample drawing below)
- ☐ The drawing shows product formation and conservation of charge.
 - 2 Ag^+ ions and 2 Al^{3+} ions in the beaker on the right (see sample drawing below)
- ☐ The drawing shows product formation and correct phases of matter for all species.
 - 6 Ag atoms that are solid and 2 Al^{3+} ions that are aqueous in the beaker on the right



Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 1: Galvanic Cell



0	1
---	---

The response provides the correct calculated value.

- $E^{\circ} = 0.80\text{V} + 1.66\text{V} = 2.46\text{V}$

OR

- $E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} - E_{\text{ox}}^{\circ} = 0.80\text{V} - (-1.66\text{V}) = 2.46\text{V}$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

- Negative. The reaction has a positive value of E° , indicating that it is thermodynamically favorable and would therefore have a negative value of ΔG° . ($\Delta G^{\circ} = -nFE^{\circ}$)

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

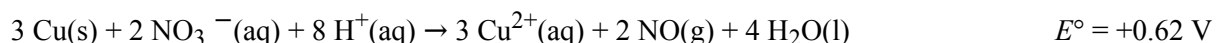
- Zero. The observation that the reaction stops progressing implies that $E_{\text{cell}} = 0$, indicating that there is no longer a driving force for the reaction.

OR

- Zero. The observation that reaction stops progressing implies that equilibrium is established, and $\Delta G = 0$ at equilibrium.

Chapter 12 Part 1: Galvanic Cell

6. Answer the following questions related to chemical reactions involving nitrogen monoxide, NO(g). The reaction between solid copper and nitric acid to form copper(II) ion, nitrogen monoxide gas, and water is represented by the following equation.

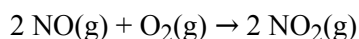


- (a) Using the information above and in the table below, calculate the standard reduction potential, E° , for the reduction of NO_3^- in acidic solution.

Half-Reaction	Standard Reduction Potential, E°
$\text{Cu}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Cu(s)}$	+0.34 V
$\text{NO}_3^-(\text{aq}) + 4 \text{ H}^+(\text{aq}) + 3 e^- \rightarrow \text{NO(g)} + 2 \text{ H}_2\text{O(l)}$	?

- (b) Calculate the value of the standard free energy change, ΔG° , for the overall reaction between solid copper and nitric acid.
- (c) Predict whether the value of the standard entropy change, ΔS° , for the overall reaction is greater than 0, less than 0, or equal to 0. Justify your prediction.

Nitrogen monoxide gas, a product of the reaction above, can react with oxygen to produce nitrogen dioxide gas, as represented below.



A rate study of the reaction yielded the data recorded in the table below.

Experiment	Initial Concentration of NO (mol L ⁻¹)	Initial Concentration of O ₂ (mol L ⁻¹)	Initial Rate of Formation of NO ₂ (mol L ⁻¹ s ⁻¹)
1	0.0200	0.0300	8.52×10^{-2}
2	0.0200	0.0900	2.56×10^{-1}
3	0.0600	0.0300	7.67×10^{-1}

- (d) Determine the order of the reaction with respect to each of the following reactants. Give details of your reasoning, clearly explaining or showing how you arrived at your answers.

- (i) NO
- (ii) O₂

- (e) Write the expression for the rate law for the reaction as determined from the experimental data.

Chapter 12 Part 1: Galvanic Cell

(f) Determine the value of the rate constant for the reaction, clearly indicating the units.

Part A

1 point is earned for the correct calculation of the standard reduction potential.

$$E_{\text{rxn}}^{\circ} = E_{\text{NO}_3^-}^{\circ} - E_{\text{Cu}^{2+}}^{\circ} = E_{\text{NO}_3^-}^{\circ} - 0.34 \text{ V} = 0.62 \text{ V}$$

$$\Rightarrow E_{\text{NO}_3^-}^{\circ} = 0.62 \text{ V} + 0.34 \text{ V} = 0.96 \text{ V}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct calculation of the standard reduction potential.

$$E_{\text{rxn}}^{\circ} = E_{\text{NO}_3^-}^{\circ} - E_{\text{Cu}^{2+}}^{\circ} = E_{\text{NO}_3^-}^{\circ} - 0.34 \text{ V} = 0.62 \text{ V}$$

$$\Rightarrow E_{\text{NO}_3^-}^{\circ} = 0.62 \text{ V} + 0.34 \text{ V} = 0.96 \text{ V}$$

Part B

1 point is earned for the correct value of n, the number of moles of electrons

1 point is earned for calculating the correct value of ΔG° , with correct sign and consistent units.

$$\Delta G^{\circ} = -nFE^{\circ} = -(6)(96,500 \text{ C mol}^{-1})(0.62 \text{ V})$$

$$= -360,000 \text{ J mol}^{-1} = -360 \text{ kJ mol}^{-1}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct value of n, the number of moles of electrons

1 point is earned for calculating the correct value of ΔG° , with correct sign and consistent units.

$$\Delta G^{\circ} = -nFE^{\circ} = -(6)(96,500 \text{ C mol}^{-1})(0.62 \text{ V})$$

$$= -360,000 \text{ J mol}^{-1} = -360 \text{ kJ mol}^{-1}$$

Part C

Chapter 12 Part 1: Galvanic Cell

1 point is earned for the correct answer with a justification that is based on the gaseous state of one of the products.

$\Delta S^\circ > 0$. Even though there is a loss of 7 moles of ions in solution, the value of ΔS° for the overall reaction will be greater than zero because two moles of NO gas will be produced (there are no gaseous reactants).



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with a justification that is based on the gaseous state of one of the products.

$\Delta S^\circ > 0$. Even though there is a loss of 7 moles of ions in solution, the value of ΔS° for the overall reaction will be greater than zero because two moles of NO gas will be produced (there are no gaseous reactants).

Part D (i)

1 point is earned for the correct answer with justification.

Comparing experiments 1 and 3, the tripling of the initial concentration of NO while the initial concentration of oxygen remained constant at $0.0300 \text{ mol L}^{-1}$ resulted in a nine-fold increase in the initial rate of formation of NO₂. Since $9 = 3^2$, the reaction is second order with respect to NO.



0	1
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The student response earns one of the following points:

1 point is earned for the correct answer with justification.

Comparing experiments 1 and 3, the tripling of the initial concentration of NO while the initial concentration of oxygen remained constant at $0.0300 \text{ mol L}^{-1}$ resulted in a nine-fold increase in the initial rate of formation of NO₂. Since $9 = 3^2$, the reaction is second order with respect to NO.

Part D (ii)

1 point is earned for the correct answer with justification.

Comparing experiments 1 and 2, the tripling of the initial concentration of O₂ while the initial concentration of NO remained constant at $0.0200 \text{ mol L}^{-1}$ resulted in a tripling in the initial rate of formation of NO₂. Since $3 = 3^1$, the reaction is first order with respect to O₂.



0	1
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Chapter 12 Part 1: Galvanic Cell

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

Comparing experiments 1 and 2, the tripling of the initial concentration of O₂ while the initial concentration of NO remained constant at 0.0200 mol L⁻¹ resulted in a tripling in the initial rate of formation of NO₂. Since 3 = 3¹, the reaction is first order with respect to O₂.

Part E

1 point is earned for the correct expression for the rate law.

$$\text{rate} = k[\text{NO}]^2[\text{O}_2]$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct expression for the rate law.

$$\text{rate} = k[\text{NO}]^2[\text{O}_2]$$

Part F

1 point is earned for calculating the correct value of the rate constant.

1 point is earned for including the correct units

Because the coefficient for NO₂ in the balanced equation is 2, the rate of the reaction is defined as 1/2 the rate of the appearance of NO₂.

$$\text{From part (e) above, } k = \frac{\text{reaction rate}}{[\text{NO}]^2[\text{O}_2]}$$

$$= \frac{(\frac{1}{2})(\text{rate of formation of NO}_2)}{[\text{NO}]^2[\text{O}_2]}$$

Substituting data from experiment 1,

$$k = \frac{(\frac{1}{2})(8.52 \times 10^{-2} \text{ mol L}^{-2} \text{ s}^{-1})}{[0.0200 \text{ mol L}^{-1}]^2 [0.0300 \text{ mol L}^{-1}]}$$

$$= 3.55 \times 10^3 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$$

Note: a rate constant value of $7.10 \times 10^3 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$ earns the point if the rate of reaction is assumed to be the same as the rate of formation of NO₂.

Chapter 12 Part 1: Galvanic Cell



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for calculating the correct value of the rate constant.

1 point is earned for including the correct units

Because the coefficient for NO₂ in the balanced equation is 2, the rate of the reaction is defined as 1/2 the rate of the appearance of NO₂.

From part (e) above, $k = \frac{\text{reaction rate}}{[\text{NO}]^2[\text{O}_2]}$

$$= \frac{\left(\frac{1}{2}\right)(\text{rate of formation of NO}_2)}{[\text{NO}]^2[\text{O}_2]}$$

Substituting data from experiment 1,

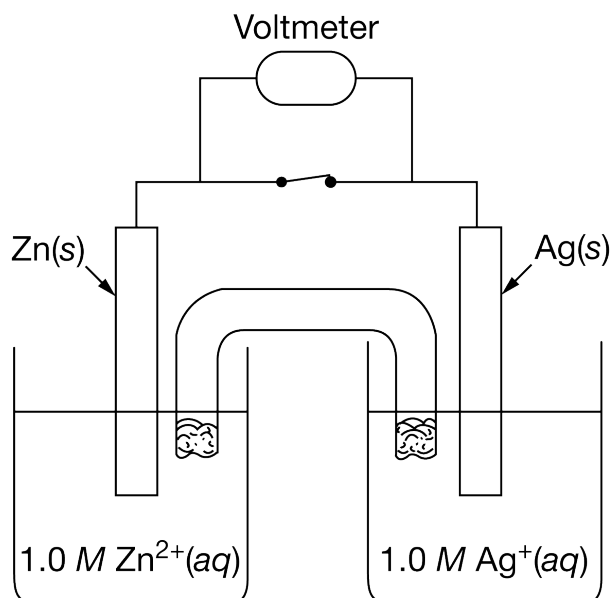
$$k = \frac{\left(\frac{1}{2}\right)(8.52 \times 10^{-2} \text{ mol L}^{-2} \text{ s}^{-1})}{[0.0200 \text{ mol L}^{-1}]^2 [0.0300 \text{ mol L}^{-1}]}$$

$$= 3.55 \times 10^3 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$$

Note: a rate constant value of $7.10 \times 10^3 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$ earns the point if the rate of reaction is assumed to be the same as the rate of formation of NO₂.

Chapter 12 Part 1: Galvanic Cell

7.



Which of the following substances would be the best choice for filling the salt bridge in the galvanic cell shown above, and why?

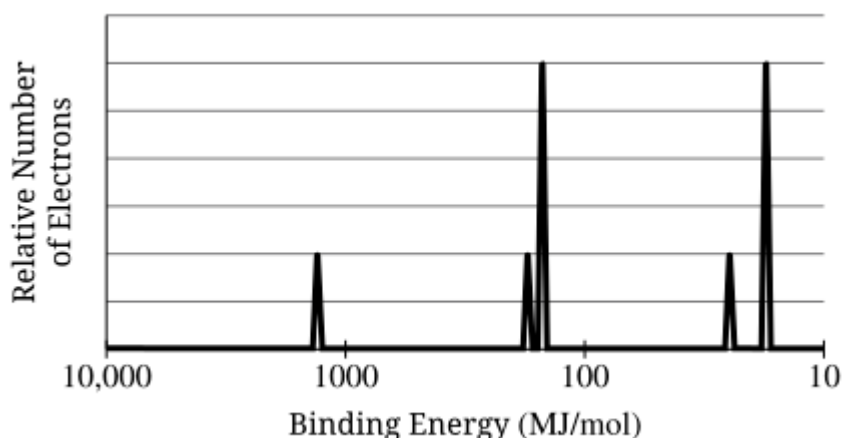
- (A) Distilled water, because it would avoid contaminating the solutions in the half-cells.
- (B) $1.0\text{ M HCl}(\text{aq})$, because it would acidify the solutions and catalyze the overall cell reaction.
- (C) $1.0\text{ M CH}_3\text{OH}(\text{aq})$, because it would increase the pH of the solutions and catalyze the overall cell reaction.
- (D) $1.0\text{ M KNO}_3(\text{aq})$, because the mobile ions would allow charge to flow between the half-cells. ✓

Chapter 12 Part 1: Galvanic Cell

8. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

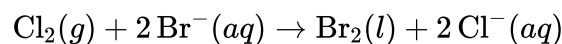
Bromine is the only nonmetal that exists as a liquid at room temperature and 1 atm.

- (a) Write the complete ground-state electron configuration of a bromine atom.
- (b) Following is a partial photoelectron spectrum (PES) showing only the core electrons in bromine.



- (i) Circle the peak in the spectrum that corresponds to the 3s electrons of bromine.
- (ii) Both bromine and krypton have 18 core electrons. Would the peaks for the core electrons in the PES of krypton be shifted to the right, shifted to the left, or remain unchanged as compared to those in the PES of bromine? Explain your reasoning.

Elemental bromine, $\text{Br}_2(l)$, is produced in the reaction represented by the following chemical equation.



- (c) A student claims that elemental chlorine, $\text{Cl}_2(g)$, is reduced in this reaction. Do you agree or disagree with the student's claim? Justify your answer in terms of oxidation numbers.

The overall reaction, related half-cell reactions, and standard reduction potentials are given in the following table.

Chapter 12 Part 1: Galvanic Cell

Reaction	E° (V)
$\text{Cl}_2(g) + 2 \text{Br}^-(aq) \rightarrow \text{Br}_2(l) + 2 \text{Cl}^-(aq)$	?
$\text{Cl}_2(g) + 2 e^- \rightarrow 2 \text{Cl}^-(aq)$	+1.36
$\text{Br}_2(l) + 2 e^- \rightarrow 2 \text{Br}^-(aq)$	+1.07

(d) Calculate the standard cell potential for the overall reaction between chlorine gas and bromide ions.

(e) Calculate the value of the standard Gibbs free energy (in $\text{kJ} / \text{mol}_{\text{rxn}}$) for the reaction between chlorine gas and bromide ions.

(f) At 298K, is this reaction thermodynamically favorable? Justify your answer based on the standard change in Gibbs free energy you calculated in part (e).

(g) The standard enthalpy change of the reaction is $\Delta H_{\text{rxn}}^\circ = -91.2 \text{ kJ/mol}_{\text{rxn}}$. Calculate the standard entropy change for the reaction at 298K. Include units with your answer.

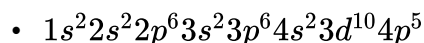
Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

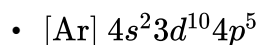


0	1
---	---

The response provides a correct answer.



OR



Part B(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

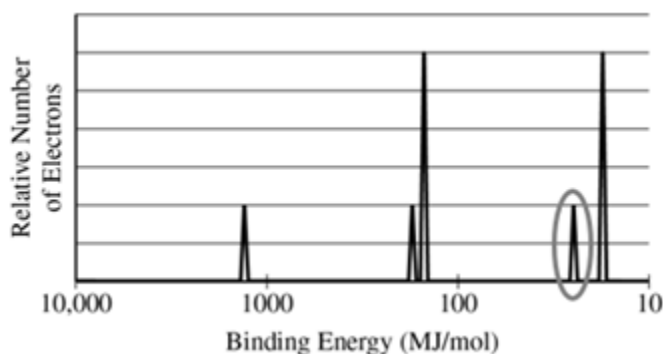


0	1
---	---

Chapter 12 Part 1: Galvanic Cell

The response provides the correct answer.

- The second peak from the right is circled.



Part B(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid explanation.

- Shifted to the left. **Kr** has a higher nuclear charge than **Br** does (36 versus 35), so the peaks for the core electrons of **Kr** would be shifted toward higher binding energies (to the left on the graph).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid explanation.

- Agree. This is a correct statement because the oxidation number of chlorine is being reduced from 0 to -1 .

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 12 Part 1: Galvanic Cell

The response provides the correct calculated value.

$$E_{cell}^{\circ} = E_{cathode}^{\circ} - E_{anode}^{\circ}$$

$$E_{cell}^{\circ} = 1.36 - 1.07 = +0.29 \text{ V}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **both** of the following criteria.

☐

The response provides the correct number of electrons.

- 2 electrons

☐

The response provides the correct calculated value, consistent with part (d).

$$\Delta G_{rxn}^{\circ} = -nFE_{cell}^{\circ} = -\left(\frac{2 \text{ mol } e^{-}}{1 \text{ mol}_{rxn}}\right)\left(\frac{96,485 \text{ coulombs}}{1 \text{ mol } e^{-}}\right)\left(\frac{0.29 \text{ J}}{1 \text{ coulomb}}\right)$$

$$\Delta G_{rxn}^{\circ} = -56,000 \text{ J/mol}_{rxn} = -56 \text{ kJ/mol}_{rxn}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification, consistent with part (e).

- Yes, this is a thermodynamically favorable reaction at 298 K because the standard Gibbs free energy change is a negative term.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

Chapter 12 Part 1: Galvanic Cell

The response meets **both** of the following criteria.

- ☐ The response provides the correct calculated value, consistent with part (e).

$$\Delta G_{rxn}^{\circ} = \Delta H_{rxn}^{\circ} - T\Delta S_{rxn}^{\circ}$$

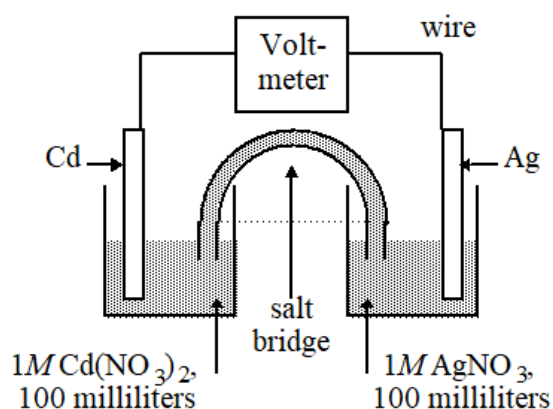
$$\Delta S_{rxn}^{\circ} = \frac{\Delta H_{rxn}^{\circ} - \Delta G_{rxn}^{\circ}}{T} = \left(\frac{-91.2 \text{ kJ/mol}_{rxn} - (-56 \text{ kJ/mol}_{rxn})}{298 \text{ K}} \right) \left(\frac{1000 \text{ J}}{\text{kJ}} \right)$$

$$\Delta S_{rxn}^{\circ} = -120 \text{ J/(K} \cdot \text{mol}_{rxn})$$

- ☐ The response provides the correct units.

$$\text{J/(K} \cdot \text{mol}_{rxn})$$

9.



The spontaneous reaction that occurs when the cell above operates is $2 \text{ Ag}^+ + \text{Cd}(s) \rightarrow 2 \text{ Ag}(s) + \text{Cd}^{2+}$

Which of the above occurs for the following circumstance?

A 50-milliliter sample of a 2-molar Cd(NO₃)₂ solution is added to the left beaker.

(A) Voltage increases.

(B) Voltage decreases but remains above zero



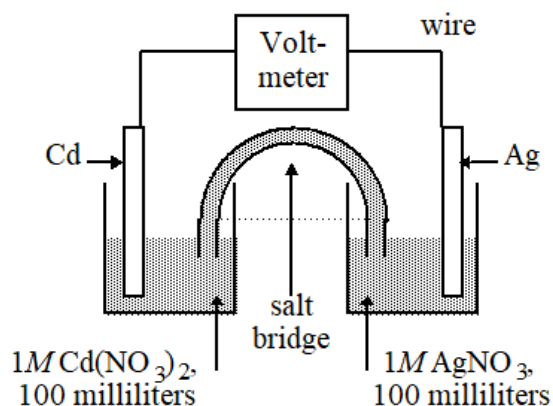
(C) Voltage becomes zero and remains at zero

(D) No change in voltage occurs

(E) Direction of voltage change cannot be predicted without additional information

Chapter 12 Part 1: Galvanic Cell

10.



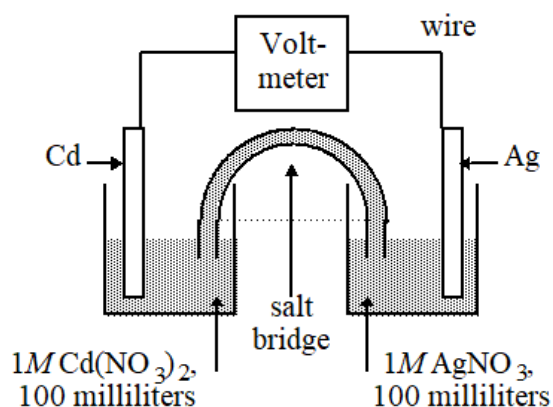
The spontaneous reaction that occurs when the cell above operates is $2 \text{Ag}^+ + \text{Cd}(s) \rightarrow 2 \text{Ag}(s) + \text{Cd}^{2+}$

Which of the above occurs for the following circumstance?

The silver electrode is made larger.

- (A) Voltage increases.
- (B) Voltage decreases but remains above zero
- (C) Voltage becomes zero and remains at zero
- (D) No change in voltage occurs ✓
- (E) Direction of voltage change cannot be predicted without additional information

11.



The spontaneous reaction that occurs when the cell above operates is $2 \text{Ag}^+ + \text{Cd}(s) \rightarrow 2 \text{Ag}(s) + \text{Cd}^{2+}$

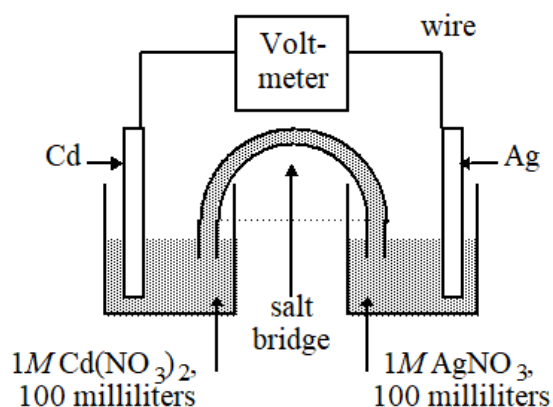
Which of the above occurs for the following circumstance?

The salt bridge is replaced by a platinum wire.

Chapter 12 Part 1: Galvanic Cell

- (A) Voltage increases.
- (B) Voltage decreases but remains above zero
- (C) Voltage becomes zero and remains at zero ✓
- (D) No change in voltage occurs
- (E) Direction of voltage change cannot be predicted without additional information

12.



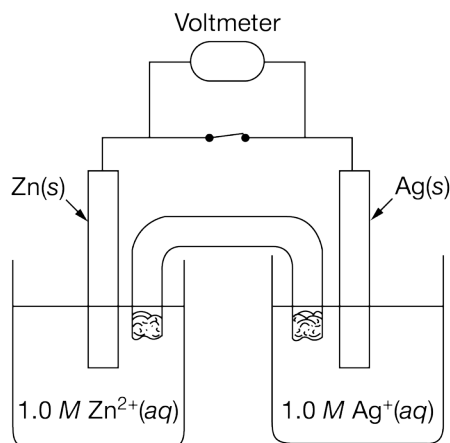
The spontaneous reaction that occurs when the cell above operates is $2 \text{Ag}^+ + \text{Cd}(s) \rightarrow 2 \text{Ag}(s) + \text{Cd}^{2+}$

Which of the above occurs for the following circumstance?

Current is allowed to flow for 5 minutes

- (A) Voltage increases.
- (B) Voltage decreases but remains above zero ✓
- (C) Voltage becomes zero and remains at zero
- (D) No change in voltage occurs
- (E) Direction of voltage change cannot be predicted without additional information

Chapter 12 Part 1: Galvanic Cell



Standard reduction potentials for the half-reactions associated with the electrochemical cell shown above are given in the table below.

Half-Reaction	E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Zn}^{2+}(aq) + 2e^- \rightarrow \text{Zn}(s)$	-0.76

13. What is the value of E° for the cell?

(A) 0.04 V

(B) 0.84 V

(C) 1.56 V

(D) 2.36 V



14. Which of the following is the net ionic equation for the overall reaction that occurs as the cell operates?

(A) $\text{Ag}(s) + \text{Zn}^{2+}(aq) \rightarrow \text{Ag}^{2+}(aq) + \text{Zn}(s)$

(B) $\text{Ag}^+(aq) + \text{Zn}^{2+}(aq) + 3e^- \rightarrow \text{AgZn}(s)$

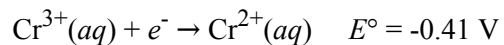
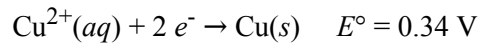
(C) $\text{Ag}^+(aq) + \text{Zn}(s) \rightarrow \text{Ag}(s) + \text{Zn}^{2+}(aq) + e^-$

(D) $2\text{Ag}^+(aq) + \text{Zn}(s) \rightarrow 2\text{Ag}(s) + \text{Zn}^{2+}(aq)$

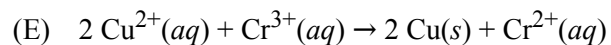
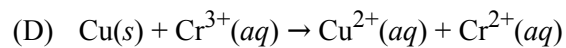
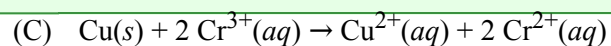
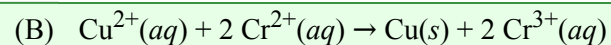
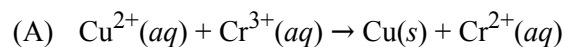


Chapter 12 Part 1: Galvanic Cell

15.

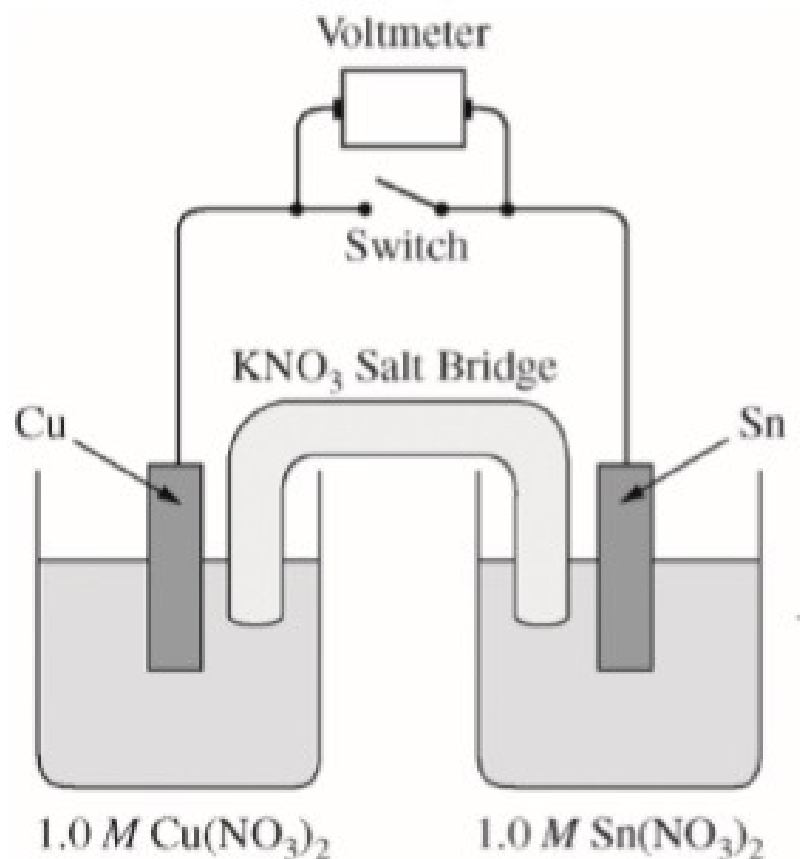


According to the half-reactions represented above, which of the following occurs in aqueous solutions under standard conditions?



Chapter 12 Part 1: Galvanic Cell

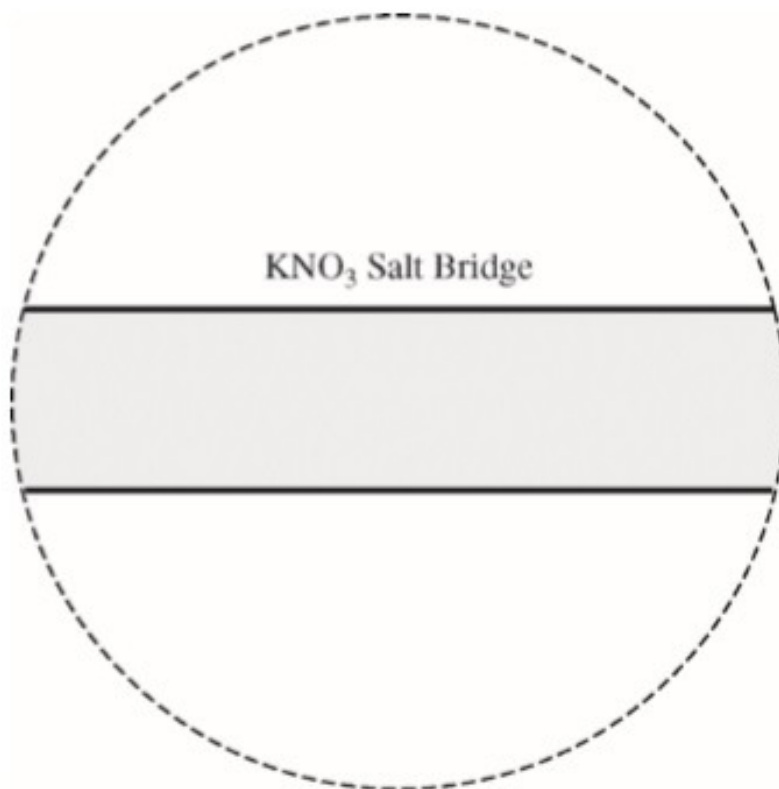
16.



A student is given a standard galvanic cell, represented above, that has a Cu electrode and a Sn electrode. As current flows through the cell, the student determines that the Cu electrode increases in mass and the Sn electrode decreases in mass.

- Identify the electrode at which oxidation is occurring. Explain your reasoning based on the student's observations.
- As the mass of the Sn electrode decreases, where does the mass go?
- In the expanded view of the center portion of the salt bridge shown in the diagram below, draw and label a particle view of what occurs in the salt bridge as the cell begins to operate. Omit solvent molecules and use arrows to show the movement of particles.

Chapter 12 Part 1: Galvanic Cell



- d. A nonstandard cell is made by replacing the 1.0 *M* solutions of $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$ in the standard cell with 0.50 *M* solutions of $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$. The volumes of solutions in the nonstandard cell are identical to those in the standard cell. $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$ in the standard cell with 0.50 *M* solutions of $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$. The volumes of solutions in the nonstandard cell are identical to those in the standard cell.
- Is the cell potential of the nonstandard cell greater than, less than, or equal to the cell potential of the standard cell? Justify your answer.
 - Both the standard and nonstandard cells can be used to power an electronic device. Would the nonstandard cell power the device for the same time, a longer time, or a shorter time as compared with the standard cell? Justify your answer.
- e. In another experiment, the student places a new Sn electrode into a fresh solution of 1.0 *M* $\text{Cu}(\text{NO}_3)_2$.

Half-Reaction	E° (V)
$\text{Cu}^+ + e^- \rightarrow \text{Cu}(s)$	0.52
$\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s)$	0.34
$\text{Sn}^{4+} + 2e^- \rightarrow \text{Sn}^{2+}$	0.15
$\text{Sn}^{2+} + 2e^- \rightarrow \text{Sn}(s)$	-0.14

Chapter 12 Part 1: Galvanic Cell

- i. Using information from the table above, write a net-ionic equation for the reaction between the Sn electrode and the $\text{Cu}(\text{NO}_3)_2$ solution that would be thermodynamically favorable. Justify that the reaction is thermodynamically favorable.
- ii. Calculate the value of ΔG° for the reaction. Include units with your answer.

Part A

1 point is earned for the correct answer with justification.

Since the Sn electrode is losing mass, Sn atoms must be forming $\text{Sn}^{2+}(\text{aq})$. This process is oxidation.

OR

because the cell operates, E° must be positive and, based on the E° values from the table, it must be Sn that is oxidized.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

Since the Sn electrode is losing mass, Sn atoms must be forming $\text{Sn}^{2+}(\text{aq})$. This process is oxidation.

OR

because the cell operates, E° must be positive and, based on the E° values from the table, it must be Sn that is oxidized.

Part B

1 point is earned for the correct answer.

The atoms on the Sn electrode are going into the solution as Sn^{2+} ions.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer.

The atoms on the Sn electrode are going into the solution as Sn^{2+} ions.

Part C

Chapter 12 Part 1: Galvanic Cell

1 point is earned for correct representation of both K^+ and NO_3^- ions. (Including free electrons loses this point.)

1 point is earned for correctly indicating the direction of movement of both ions.

The response should show at least one K^+ ion moving toward the Cu compartment on the left and at least one NO_3^- ion moving in the opposite direction.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correct representation of both K^+ and NO_3^- ions. (Including free electrons loses this point.)

1 point is earned for correctly indicating the direction of movement of both ions.

The response should show at least one K^+ ion moving toward the Cu compartment on the left and at least one NO_3^- ion moving in the opposite direction.

Part D

- i. 1 point is earned for the correct answer with justification.

It is the same. In the cell reaction $Q = \frac{[\text{Sn}^{2+}]}{[\text{Cu}^{2+}]}$, and the concentrations of Sn^{2+} and Cu^{2+} are equal to each other in both cases.

- ii. 1 point is earned for the correct answer with justification.

The nonstandard cell would power the device for a shorter time because the supply of Cu^{2+} ions will be exhausted more quickly.

OR

The nonstandard cell would power the device for a shorter time because the reaction will reach $E = 0$ more quickly.



0	1	2
---	---	---

The student response earns none of the following points:

- i. 1 point is earned for the correct answer with justification.

It is the same. In the cell reaction $Q = \frac{[\text{Sn}^{2+}]}{[\text{Cu}^{2+}]}$, and the concentrations of Sn^{2+} and Cu^{2+} are equal to each other in both cases.

Chapter 12 Part 1: Galvanic Cell

- ii. 1 point is earned for the correct answer with justification.

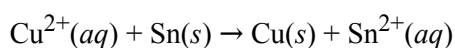
The nonstandard cell would power the device for a shorter time because the supply of Cu^{2+} ions will be exhausted more quickly.

OR

The nonstandard cell would power the device for a shorter time because the reaction will reach $E = 0$ more quickly.

Part E

- i. 1 point is earned for the correct net-ionic equation.



1 point is earned for a correct justification (unit not needed in calculation).

E° is positive ($0.34 \text{ V} + 0.14 \text{ V} = 0.48 \text{ V}$), therefore the reaction is thermodynamically favorable.

OR

The cell observations from earlier parts of the question are evidence that the Sn is oxidized and Cu is reduced, therefore E° must be positive.

- ii. 1 point is earned for the correct number of electrons.

1 point is earned for the correct answer with unit.

$$\Delta G^\circ = -nFE^\circ$$

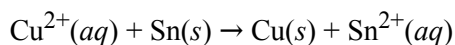
$$\Delta G = -\frac{2 \text{ mol } e^-}{\text{mol}_{rxn}} \times \frac{96,485 \text{ C}}{\text{mol } e^-} \times \frac{0.48 \text{ J}}{\text{C}} = -93,000 \text{ J/mol}_{rxn} = -93 \text{ kJ/mol}_{rxn}$$



0	1	2	3	4
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The student response earns four of the following points:

- i. 1 point is earned for the correct net-ionic equation.



1 point is earned for a correct justification (unit not needed in calculation).

E° is positive ($0.34 \text{ V} + 0.14 \text{ V} = 0.48 \text{ V}$), therefore the reaction is thermodynamically favorable.

OR

The cell observations from earlier parts of the question are evidence that the Sn is oxidized and Cu is reduced, therefore E° must be positive.

Chapter 12 Part 1: Galvanic Cell

- ii. 1 point is earned for the correct number of electrons.

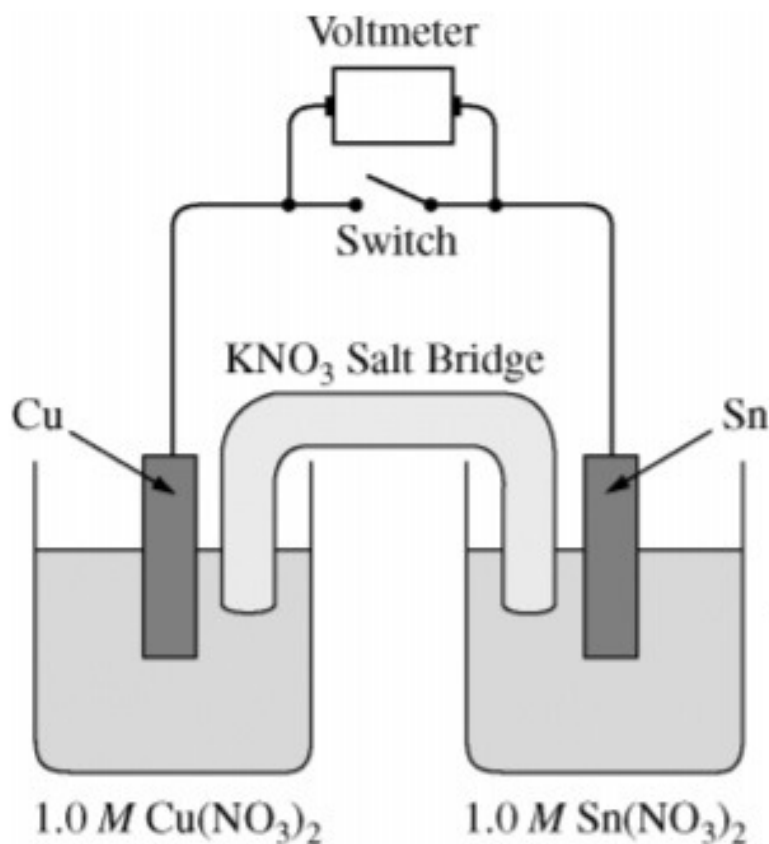
1 point is earned for the correct answer with unit.

$$\Delta G^\circ = -nFE^\circ$$

$$\Delta G = -\frac{2 \text{ mol } e^-}{\text{mol}_{rxn}} \times \frac{96,485 \text{ C}}{\text{mol } e^-} \times \frac{0.48 \text{ J}}{\text{C}} = -93,000 \text{ J/mol}_{rxn} = -93 \text{ kJ/mol}_{rxn}$$

Chapter 12 Part 1: Galvanic Cell

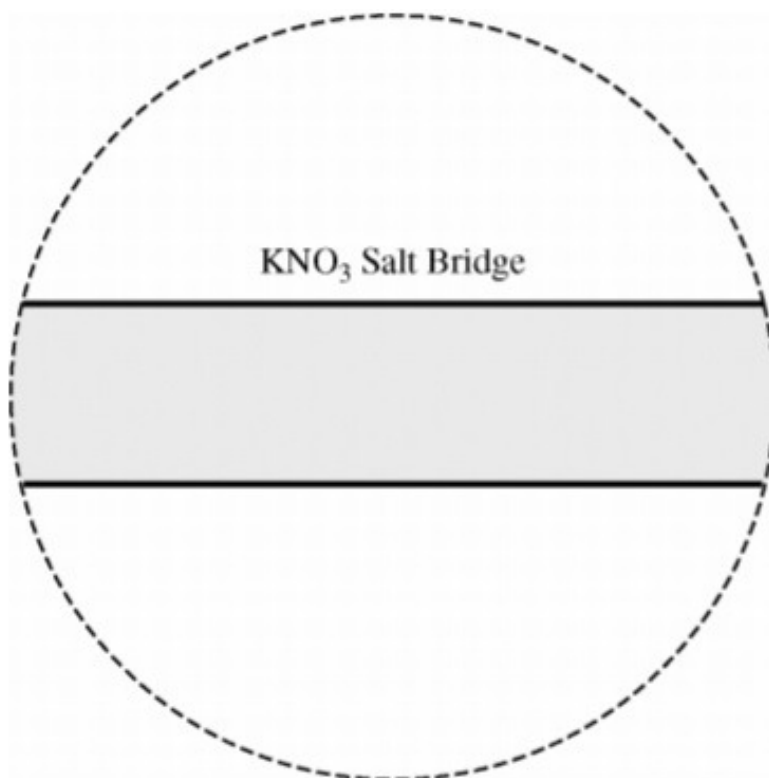
17.



A student is given a standard galvanic cell, represented above, that has a Cu electrode and a Sn electrode. As current flows through the cell, the student determines that the Cu electrode increases in mass and the Sn electrode decreases in mass.

- Identify the electrode at which oxidation is occurring. Explain your reasoning based on the student's observations.
- As the mass of the Sn electrode decreases, where does the mass go?
- In the expanded view of the center portion of the salt bridge shown in the diagram below, draw and label a particle view of what occurs in the salt bridge as the cell begins to operate. Omit solvent molecules and use arrows to show the movement of particles.

Chapter 12 Part 1: Galvanic Cell



- d. A nonstandard cell is made by replacing the 1.0 M solutions of $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$ in the standard cell with 0.50 M solutions of $\text{Cu}(\text{NO}_3)_2$ and $\text{Sn}(\text{NO}_3)_2$. The volumes of solutions in the nonstandard cell are identical to those in the standard cell.
- Is the cell potential of the nonstandard cell greater than, less than, or equal to the cell potential of the standard cell? Justify your answer.
 - Both the standard and nonstandard cells can be used to power an electronic device. Would the nonstandard cell power the device for the same time, a longer time, or a shorter time as compared with the standard cell? Justify your answer.
- e. In another experiment, the student places a new Sn electrode into a fresh solution of 1.0 M $\text{Cu}(\text{NO}_3)_2$.

Half-Reaction	E° (V)
$\text{Cu}^+ + e^- \rightarrow \text{Cu}(s)$	0.52
$\text{Cu}^{2+} + 2 e^- \rightarrow \text{Cu}(s)$	0.34
$\text{Sn}^{4+} + 2 e^- \rightarrow \text{Sn}^{2+}$	0.15
$\text{Sn}^{2+} + 2 e^- \rightarrow \text{Sn}(s)$	-0.14

Chapter 12 Part 1: Galvanic Cell

- i. Using information from the table above, write a net-ionic equation for the reaction between the Sn electrode and the $\text{Cu}(\text{NO}_3)_2$ solution that would be thermodynamically favorable. Justify that the reaction is thermodynamically favorable.
- ii. Calculate the value of ΔG° for the reaction. Include units with your answer.

Part A

1 point is earned for the correct answer with justification.

Since the Sn electrode is losing mass, Sn atoms must be forming $\text{Sn}^{2+}(\text{aq})$. This process is oxidation.

OR

because the cell operates, E° must be positive and, based on the E° values from the table, it must be Sn that is oxidized.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

Since the Sn electrode is losing mass, Sn atoms must be forming $\text{Sn}^{2+}(\text{aq})$. This process is oxidation.

OR

because the cell operates, E° must be positive and, based on the E° values from the table, it must be Sn that is oxidized.

Part B

1 point is earned for the correct answer.

The atoms on the Sn electrode are going into the solution as Sn^{2+} ions.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer.

The atoms on the Sn electrode are going into the solution as Sn^{2+} ions.

Part C

Chapter 12 Part 1: Galvanic Cell

1 point is earned for correct representation of both K^+ and NO_3^- ions. (Including free electrons loses this point.)

1 point is earned for correctly indicating the direction of movement of both ions.

The response should show at least one K^+ ion moving toward the Cu compartment on the left and at least one NO_3^- ion moving in the opposite direction.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correct representation of both K^+ and NO_3^- ions. (Including free electrons loses this point.)

1 point is earned for correctly indicating the direction of movement of both ions.

The response should show at least one K^+ ion moving toward the Cu compartment on the left and at least one NO_3^- ion moving in the opposite direction.

Part D

- 1 point is earned for the correct answer with justification.

It is the same. In the cell reaction $Q = \frac{[Sn^{2+}]}{[Cu^{2+}]}$, and the concentrations of Sn^{2+} and Cu^{2+} are equal to each other in both cases

- 1 point is earned for the correct answer with justification.

The nonstandard cell would power the device for a shorter time because the supply of Cu^{2+} ions will be exhausted more quickly.

OR

The nonstandard cell would power the device for a shorter time because the reaction will reach $E = 0$ more quickly.



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for the correct answer with justification.

It is the same. In the cell reaction $Q = \frac{[Sn^{2+}]}{[Cu^{2+}]}$, and the concentrations of Sn^{2+} and Cu^{2+} are equal to each other in both cases

Chapter 12 Part 1: Galvanic Cell

- 1 point is earned for the correct answer with justification.

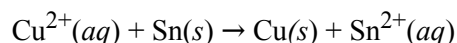
The nonstandard cell would power the device for a shorter time because the supply of Cu^{2+} ions will be exhausted more quickly.

OR

The nonstandard cell would power the device for a shorter time because the reaction will reach $E = 0$ more quickly.

Part E

- 1 point is earned for the correct net-ionic equation.



1 point is earned for a correct justification (unit not needed in calculation).

E° is positive ($0.34 \text{ V} + 0.14 \text{ V} = 0.48 \text{ V}$), therefore the reaction is thermodynamically favorable.

OR

The cell observations from earlier parts of the question are evidence that the Sn is oxidized and Cu is reduced, therefore E° must be positive.

- 1 point is earned for the correct number of electrons.

1 point is earned for the correct answer with unit.

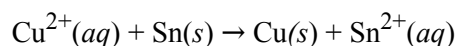
$$\Delta G^\circ = -nFE^\circ$$

$$\Delta G^\circ = -\frac{2 \text{ mol } e^-}{\text{mol}_{\text{rxn}}} \times \frac{96,485 \text{ C}}{\text{mol } e^-} \times \frac{0.48 \text{ J}}{\text{C}} = -93,000 \text{ J/mol}_{\text{rxn}} = -93 \text{ kJ/mol}_{\text{rxn}}$$



0	1	2	3	4
---	---	---	---	---

- 1 point is earned for the correct net-ionic equation.



1 point is earned for a correct justification (unit not needed in calculation).

E° is positive ($0.34 \text{ V} + 0.14 \text{ V} = 0.48 \text{ V}$), therefore the reaction is thermodynamically favorable.

OR

The cell observations from earlier parts of the question are evidence that the Sn is oxidized and Cu is reduced, therefore E° must be positive.

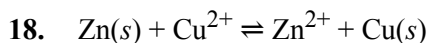
- 1 point is earned for the correct number of electrons.

Chapter 12 Part 1: Galvanic Cell

1 point is earned for the correct answer with unit.

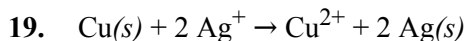
$$\Delta G^\circ = -nFE^\circ$$

$$\Delta G^\circ = -\frac{2 \text{ mol } e^-}{\text{mol}_{\text{rxn}}} \times \frac{96,485 \text{ C}}{\text{mol } e^-} \times \frac{0.48 \text{ J}}{\text{C}} = -93,000 \text{ J/mol}_{\text{rxn}} = -93 \text{ kJ/mol}_{\text{rxn}}$$



An electrolytic cell based on the reaction represented above was constructed from zinc and copper half-cells. The observed voltage was found to be 1.00 volt instead of the standard cell potential, E° , of 1.10 volts. Which of the following could correctly account for this observation?

- (A) The copper electrode was larger than the zinc electrode.
- (B) The Zn^{2+} electrolyte was $\text{Zn}(\text{NO}_3)_2$, while the Cu^{2+} electrolyte was CuSO_4 .
- (C) The Zn^{2+} solution was more concentrated than the Cu^{2+} solution. ✓
- (D) The solutions in the half-cells had different volumes.
- (E) The salt bridge contained KCl as the electrolyte.

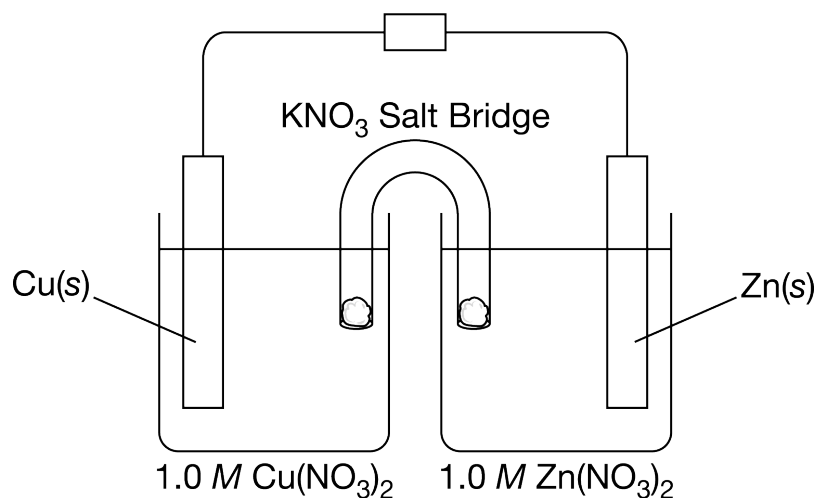


If the equilibrium constant for the reaction above is 3.7×10^{15} , which of the following correctly describes the standard voltage, E° , and the standard free energy change, ΔG° , for this reaction?

- (A) E° is positive and ΔG° is negative. ✓
- (B) E° is negative and ΔG° is positive.
- (C) E° and ΔG° are both positive.
- (D) E° and ΔG° are both negative.
- (E) E° and ΔG° are both zero.

Chapter 12 Part 1: Galvanic Cell

20. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

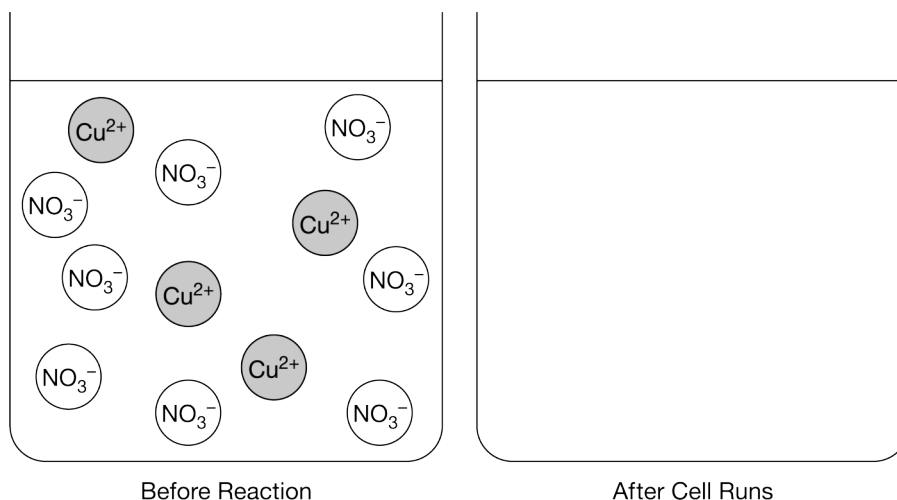


Answer the following questions related to the standard galvanic cell represented in the diagram above. The reduction half-reactions for Cu^{2+} and Zn^{2+} are given in the table below.

Half-Reaction	E° (V)
$\text{Cu}^{2+} + 2 e^- \rightarrow \text{Cu}(s)$	0.34
$\text{Zn}^{2+} + 2 e^- \rightarrow \text{Zn}(s)$	-0.76

- (a) Write the net-ionic equation for the overall reaction that takes place as the cell operates.
- (b) Calculate the value of E° for the cell.
- (c) Calculate ΔG° , in kJ/mol_{rxn} , for the cell.
- (d) The following diagram on the left represents the copper nitrate solution in the half-cell before the galvanic cell starts to operate. In the following diagram on the right, draw a representation of the solution in the same half-cell after the galvanic cell has been operating for some time and is still producing a voltage.

Chapter 12 Part 1: Galvanic Cell



(e) Assume that the mass of the Cu electrode changes by x grams in a certain period of time. Write a mathematical expression for the change in mass of the Zn electrode during the same period of time.

(f) For the standard cell, the Cu^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Cu}(\text{NO}_3)_2$ and the Zn^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Zn}(\text{NO}_3)_2$. The experiment was repeated, but this time the Cu^{2+} half-cell was made with 0.50 L of 2.0 M $\text{Cu}(\text{NO}_3)_2$ and the Zn^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Zn}(\text{NO}_3)_2$. Is the cell potential for the nonstandard cell greater than, less than, or equal to the value calculated in part (b)? Justify your answer.

(g) If the experiment is repeated with the Cu electrode replaced with a Cu electrode with twice the mass, will the initial cell voltage be greater than, less than, or equal to the initial cell voltage in the original experiment? Justify your answer.

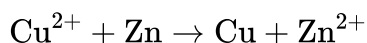
Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives the following equation:

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 1: Galvanic Cell



0	1
---	---

The response gives the following calculation:

$$E_{cell}^{\circ} = 0.34 \text{ V} - (-0.76 \text{ V}) = 1.10 \text{ V}$$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If the student makes an error, the student can still earn credit for following steps using incorrect values.



0	1	2
---	---	---

The response meets both of the following criteria:

- ☐ The response gives the correct setup.

$$\Delta G^{\circ} = -nFE_{cell}^{\circ} = -(2 \text{ mol})(96,485 \frac{\text{J}}{\text{V mol}})(1.10 \text{ V})$$
- ☐ The response shows the correct result of the calculation.

$$\Delta G^{\circ} = -212,000 \text{ J/mol}_{rxn} = -212 \text{ kJ/mol}_{rxn}$$

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The diagram should be consistent with the reaction written in part (a). If the equation in part (a) is reversed, the diagram should contain five, six, or seven Cu^{2+} ions.



0	1	2
---	---	---

The response meets both of the following criteria:

- ☐ The diagram contains one, two, or three Cu^{2+} ions.
- ☐ There is a charge balance in the diagram (the student can add K^{+} ions from the salt bridge or put in fewer NO_3^{-} ions).

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 1: Galvanic Cell



0	1
---	---

The response indicates the following:

$$x \text{ g of Cu} \times \frac{1 \text{ mol Cu}}{63.55 \text{ g Cu}} \times \frac{1 \text{ mol Zn}}{1 \text{ mol Cu}} \times \frac{65.38 \text{ g Zn}}{1 \text{ mol Zn}} = \text{g of Zn}$$

$$\text{Change in mass of Zn electrode} = -\frac{65.38}{63.55} x \text{ g}$$

Part (f)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets both of the following criteria:

☐

The response gives the following justification:

$$E = E^\circ - \frac{RT}{nF} \ln Q = E^\circ - \frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = E^\circ - \frac{RT}{nF} \ln \frac{1}{2}$$

The log of a number less than one is negative, so for the nonstandard cell a negative number is subtracted from the standard potential: If $E = E^\circ - \frac{RT}{nF}$ (negative number), then E will be greater than E° .

☐

The response gives the following conclusion: The cell potential for the nonstandard cell is greater than that for the standard cell (or E is greater than E°).

Part (g)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response indicates the following:

The initial voltage when the mass of the Cu electrode is twice the original mass will be equal to the voltage in the original experiment. The mass and concentration of Cu(s) are not in the equation for the cell potential and do not affect it.

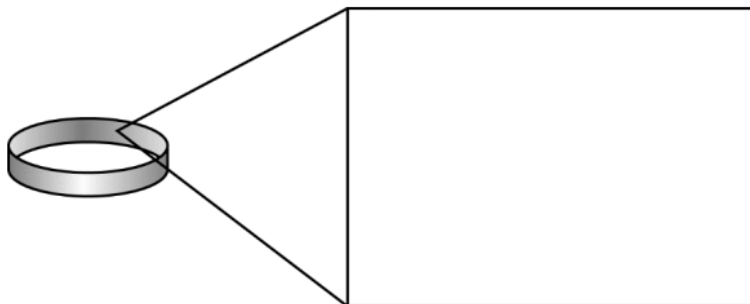
Chapter 12 Part 1: Galvanic Cell

21. **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

Element	Atomic Radius (pm)
Au	135
Pd	140

White gold is a common alloy of gold and palladium that is often used in jewelry. The atomic radii of the metals are given in the table above.

- (a) A particular ring is made from an alloy that is 75 mole percent gold and 25 mole percent palladium. Using the box below, draw a particle-level diagram of the solid alloy consisting of 12 atoms with a representative proportion of atom types. Your diagram should clearly indicate whether the alloy is interstitial or substitutional. Use empty circles for gold and shaded circles for palladium.



White-gold jewelry is often coated with rhodium to modify the color and durability.

- (b) A student hypothesizes that placing the ring in a 1.0 M solution of $\text{Rh}(\text{NO}_3)_3(\text{aq})$ will result in a reaction in which Rh metal is deposited on the white-gold ring. Based on the information in the table below, calculate E° for the reactions that may occur between ions in the solution and atoms in the ring, and indicate whether or not the student's hypothesis is correct.

Half-Reaction	E° (V)
$\text{Au}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Au}(\text{s})$	1.50
$\text{Pd}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Pd}(\text{s})$	0.92
$\text{Rh}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Rh}(\text{s})$	0.76

Chapter 12 Part 1: Galvanic Cell**Part A****2 point(s) maximum**

1 point is earned for the correct number of each type of circle.

1 point is earned for showing an acceptable arrangement (i.e., regular array of circles of about the same size).

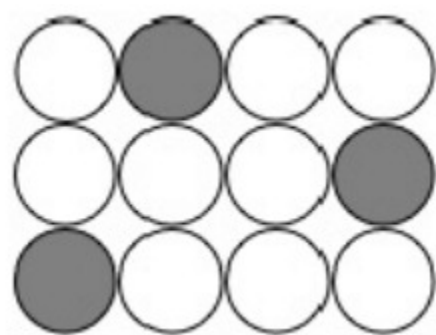


Diagram should show 12 circles in a regular array, 9 empty and 3 shaded.



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the correct number of each type of circle.

1 point is earned for showing an acceptable arrangement (i.e., regular array of circles of about the same size).

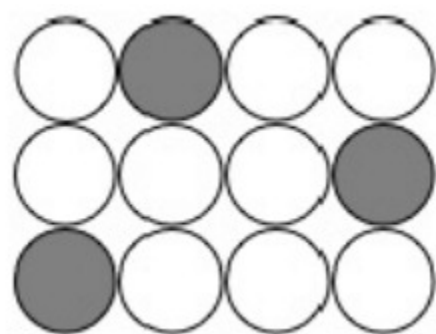
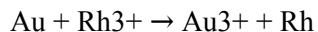


Diagram should show 12 circles in a regular array, 9 empty and 3 shaded.

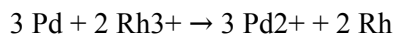
Part B**2 point(s) maximum**

Chapter 12 Part 1: Galvanic Cell

1 point is earned for the E° calculations.



$$0.76 \text{ V} - 1.50 \text{ V} = -0.74 \text{ V}$$



$$0.76 \text{ V} - 0.92 \text{ V} = -0.16 \text{ V}$$

1 point is earned for the claim

The student's hypothesis is incorrect. (Both voltages are negative, so the reactions are not thermodynamically favorable under these conditions.)

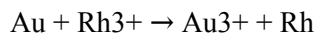


0	1	2
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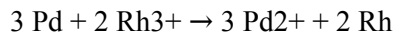
Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the E° calculations.



$$0.76 \text{ V} - 1.50 \text{ V} = -0.74 \text{ V}$$



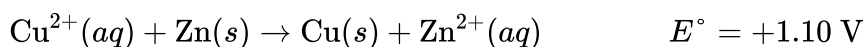
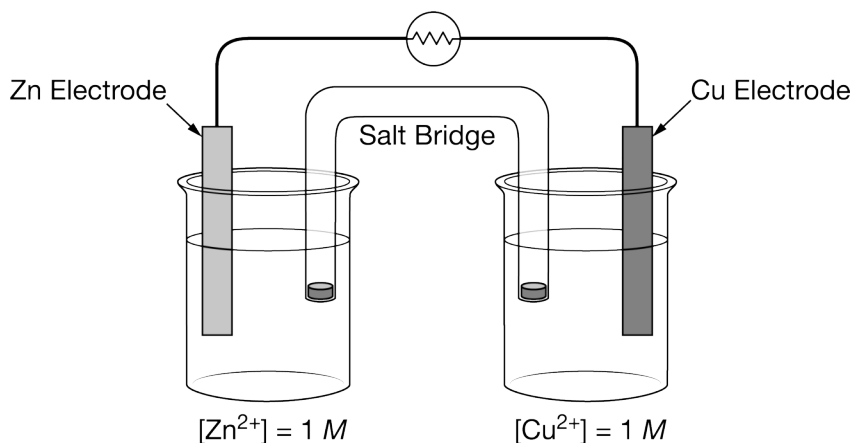
$$0.76 \text{ V} - 0.92 \text{ V} = -0.16 \text{ V}$$

1 point is earned for the claim

The student's hypothesis is incorrect. (Both voltages are negative, so the reactions are not thermodynamically favorable under these conditions.)

Chapter 12 Part 1: Galvanic Cell

22.



The galvanic cell illustrated above generates a potential of +1.10 V. For the construction of a second galvanic cell (not shown), only one modification was made: the Cu electrode has double the mass of the Cu electrode in the first cell. Which of the following correctly compares the initial E° for the second cell to that of first cell at 298 K, and why?

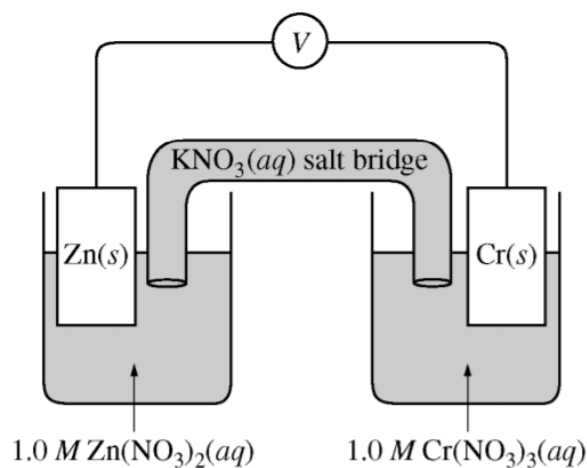
- (A) The initial E° for the second cell is twice that of the first cell, because a larger amount of $\text{Cu}(\text{s})$ can be oxidized to Cu^{2+} .
- (B) The initial E° for the second cell is twice that of the first cell, because more Cu^{2+} ions can be deposited on the solid Cu electrode.
- (C) The initial E° for the second cell is half that of the first cell, because the greater amount $\text{Cu}(\text{s})$ in the half-cell inhibits the formation of more $\text{Cu}(\text{s})$.
- (D) The initial E° for the second cell is the same as for the first cell, because the overall chemical reaction that occurs in the cell does not change. ✓

23. When a magnesium wire is dipped into a solution of lead (II) nitrate, a black deposit forms on the wire. Which of the following can be concluded from this observation?

- (A) The standard reduction potential, E° , for $\text{Pb}^{2+}(\text{aq})$ is greater than that for $\text{Mg}^{2+}(\text{aq})$. ✓
- (B) $\text{Mg}(\text{s})$ is less easily oxidized than $\text{Pb}(\text{s})$.
- (C) An external source of potential must have been supplied.
- (D) The magnesium wire will be the cathode of a Mg/Pb cell.
- (E) $\text{Pb}(\text{s})$ can spontaneously displace $\text{Mg}^{2+}(\text{aq})$ from solution.

Chapter 12 Part 1: Galvanic Cell

24.



The standard reduction potentials for the half-reactions related to the galvanic cell represented above are listed in the table below.

Half-Reaction	E° (V)
$\text{Cr}^{3+}(aq) + 3 e^- \rightarrow \text{Cr}(s)$	-0.74
$\text{Zn}^{2+}(aq) + 2 e^- \rightarrow \text{Zn}(s)$	-0.76

Which of the following gives the value of E°_{cell} for the cell?

- (A) $E^\circ_{\text{cell}} = -1.50\text{ V}$
(B) $E^\circ_{\text{cell}} = -0.80\text{ V}$
(C) $E^\circ_{\text{cell}} = -0.02\text{ V}$
(D) $E^\circ_{\text{cell}} = +0.02\text{ V}$



Chapter 12 Part 1: Galvanic Cell

25. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Half-Reaction	E° (V)
$\text{O}_2 + 4 \text{H}^+ + 4 e^- \rightarrow 2 \text{H}_2\text{O}$	+0.815
$\text{NAD}^+ + \text{H}^+ + 2 e^- \rightarrow \text{NADH}$	−0.320

In human cells a compound known as NADH is involved in a reaction that generates an electrical potential. The reaction is based on the two half-reactions shown in the table above. The balanced equation for the overall reaction is shown below.



- (a) How many moles of electrons are transferred for each mole of O_2 that reacts? Justify your answer in terms of the balanced equation and the change in oxidation number of oxygen.
- (b) Calculate the electrochemical potential of the overall reaction.
- (c) Calculate the value of ΔG° for the overall reaction.
- (d) When the concentration of NADH increases while the concentration of NAD^+ decreases, will the electrochemical potential increase, decrease, or remain the same? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response indicates that four moles of electrons are needed to convert the two moles of oxygen atoms in one mole of O_2 from the zero oxidation state to the -2 oxidation state.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 12 Part 1: Galvanic Cell

The response gives a calculation equivalent to the following.

$$E_{cell}^{\circ} = +0.815 \text{ V} - (-0.320 \text{ V}) = +1.135 \text{ V}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives a calculation equivalent to the following.

$$\begin{aligned}\Delta G^{\circ} &= -nFE^{\circ} \\ &= -\frac{4 \text{ mol } e^{-}}{1 \text{ mol}_{rxn}} \times \frac{96,485 \text{ C}}{1 \text{ mol } e^{-}} \times \frac{1.135 \text{ J}}{1 \text{ C}} \\ &= -4.380 \times 10^5 \text{ J/mol}_{rxn} = -438.0 \text{ kJ/mol}_{rxn}\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



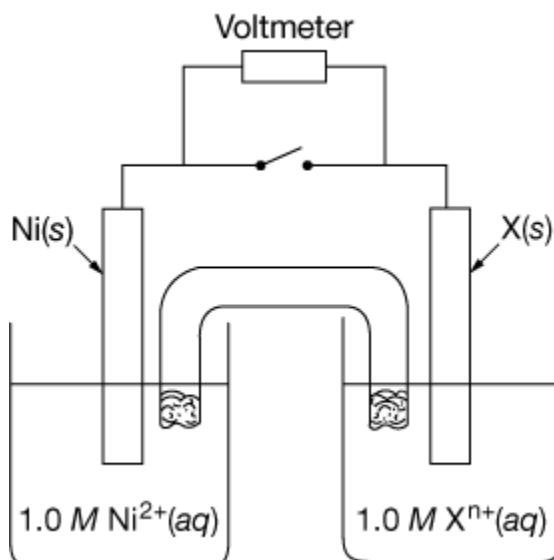
0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the electrochemical potential will increase.
- ☐ The response includes a justification such as the following.
The highly negative ΔG° indicates that the value of K for this reaction is very large, so decreasing Q causes the system to deviate further from equilibrium, thus increasing the magnitude of the electrochemical potential.

Chapter 12 Part 1: Galvanic Cell

26.



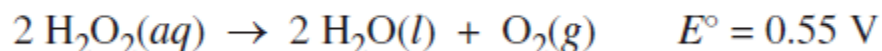
In the galvanic cell shown above, the voltmeter connected across the electrodes shows a voltage of 0.52 V. Based on the standard reduction potentials given in the table below, which of the following is most likely the identity of the unknown element, X?

Half-Reaction	E° (V)
$\text{Rh}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Rh}(\text{s})$	0.76
$\text{Bi}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Bi}(\text{s})$	0.31
$\text{In}^+(\text{aq}) + e^- \rightarrow \text{In}(\text{s})$	-0.14
$\text{Ni}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Ni}(\text{s})$	-0.24
$\text{Zn}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Zn}(\text{s})$	-0.76

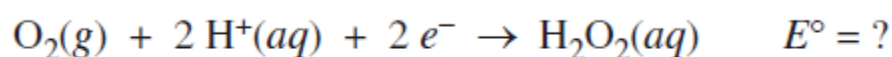
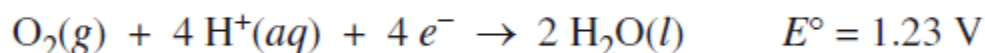
- (A) Rh
(B) Bi
(C) In
(D) Zn



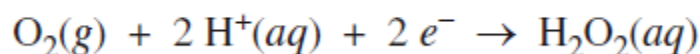
Chapter 12 Part 1: Galvanic Cell



The equation and standard cell potential for the decomposition of $\text{H}_2\text{O}_2(aq)$ in acidic solution at 25°C is given above. The reduction half-reactions for the process are listed below.



27.



What is the standard reduction potential for the half-reaction represented above?

(A) -1.78 V (B) -0.68 V (C) $+0.68 \text{ V}$ (D) $+1.78 \text{ V}$ 28. Which of the following is true for the decomposition of $\text{H}_2\text{O}_2(aq)$?(A) $\Delta G^\circ > 0$ and $K_{eq} > 1$ (B) $\Delta G^\circ > 0$ and $K_{eq} < 1$ (C) $\Delta G^\circ < 0$ and $K_{eq} > 1$ (D) $\Delta G^\circ < 0$ and $K_{eq} < 1$

Chapter 12 Part 1: Galvanic Cell

29.

Half-cell 1: strip of $\text{Al}(s)$ in $1.00\text{ M Al(NO}_3)_3(aq)$ Half-cell 2: strip of $\text{Cu}(s)$ in $1.00\text{ M Cu(NO}_3)_2(aq)$ Half-cell 3: strip of $\text{Fe}(s)$ in $1.00\text{ M Fe(NO}_3)_2(aq)$

Galvanic Cell	Half-cells	Cell Reaction	E°_{cell} (V)
X	1 and 2	$2\text{ Al}(s) + 3\text{ Cu}^{2+}(aq) \rightarrow 2\text{ Al}^{3+}(aq) + 3\text{ Cu}(s)$	2.00
Y	1 and 3	$2\text{ Al}(s) + 3\text{ Fe}^{2+}(aq) \rightarrow 2\text{ Al}^{3+}(aq) + 3\text{ Fe}(s)$	1.22
Z	2 and 3	$\text{Fe}(s) + \text{Cu}^{2+}(aq) \rightarrow \text{Fe}^{2+}(aq) + \text{Cu}(s)$	?

If the half-cell containing $1.00\text{ M Fe(NO}_3)_2(aq)$ in galvanic cells Y and Z is replaced with a half-cell containing $5.00\text{ M Fe(NO}_3)_2(aq)$, what will be the effect on the cell voltage of the two galvanic cells?

- (A) The voltage will increase in both cells.
- (B) The voltage will decrease in both cells.
- (C) The voltage will increase in cell Y and decrease in cell Z. ✓
- (D) The voltage will decrease in cell Y and increase in cell Z.

30.

Half-cell 1: strip of $\text{Al}(s)$ in $1.00\text{ M Al(NO}_3)_3(aq)$ Half-cell 2: strip of $\text{Cu}(s)$ in $1.00\text{ M Cu(NO}_3)_2(aq)$ Half-cell 3: strip of $\text{Fe}(s)$ in $1.00\text{ M Fe(NO}_3)_2(aq)$

Galvanic Cell	Half-cells	Cell Reaction	E°_{cell} (V)
X	1 and 2	$2\text{ Al}(s) + 3\text{ Cu}^{2+}(aq) \rightarrow 2\text{ Al}^{3+}(aq) + 3\text{ Cu}(s)$	2.00
Y	1 and 3	$2\text{ Al}(s) + 3\text{ Fe}^{2+}(aq) \rightarrow 2\text{ Al}^{3+}(aq) + 3\text{ Fe}(s)$	1.22
Z	2 and 3	$\text{Fe}(s) + \text{Cu}^{2+}(aq) \rightarrow \text{Fe}^{2+}(aq) + \text{Cu}(s)$	?

In galvanic cells Y and Z, which of the following takes place in half-cell 3 ?

- (A) Reduction occurs in both cell Y and cell Z.
- (B) Oxidation occurs in both cell Y and cell Z.
- (C) Reduction occurs in cell Y, and oxidation occurs in cell Z. ✓
- (D) Oxidation occurs in cell Y, and reduction occurs in cell Z.

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31. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions relating to the chemistry of the halogens.

(a) The molecular formulas of diatomic bromine, chlorine, fluorine, and iodine are written below. Circle the formula of the molecule that has the longest bond length. Justify your choice in terms of atomic structure.

Br₂ Cl₂ F₂ I₂

A chemistry teacher wants to prepare Br₂. The teacher has access to the following three reagents: NaBr(aq), Cl₂(g), and I₂(s).

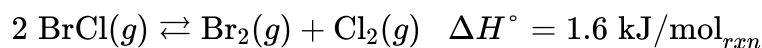
Half-Reaction	E° at 25°C (V)
$\text{Br}_2 + 2 e^- \rightarrow 2 \text{Br}^-$	1.07
$\text{Cl}_2 + 2 e^- \rightarrow 2 \text{Cl}^-$	1.36
$\text{I}_2 + 2 e^- \rightarrow 2 \text{I}^-$	0.53

(b) Using the data in the table above, write the balanced equation for the thermodynamically favorable reaction that will produce Br₂ when the teacher combines two of the reagents. Justify that the reaction is thermodynamically favorable by calculating the value of E° for the reaction.

Br₂ and Cl₂ can react to form the compound BrCl.

(c) The boiling point of Br₂ is 332 K, whereas the boiling point of BrCl is 278 K. Explain this difference in boiling point in terms of all the intermolecular forces present between molecules of each substance.

The compound BrCl can decompose into Br₂ and Cl₂, as represented by the balanced chemical equation below.



A 0.100 mole sample of pure BrCl(g) is placed in a previously evacuated, rigid 2.00 L container at 298 K. Eventually the system reaches equilibrium according to the equation above.

(d) Calculate the pressure in the container before equilibrium is established.

(e) Write the expression for the equilibrium constant, K_{eq} , for the decomposition of BrCl.

After the system has reached equilibrium, 42 percent of the original BrCl sample has decomposed.

(f) Determine the value of K_{eq} for the decomposition reaction of BrCl at 298 K.

(g) Calculate the bond energy of the Br — Cl bond, in kJ/mol, using ΔH° for the reaction

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($1.6 \text{ kJ/mol}_{\text{rxn}}$) and the information in the following table.

Bond	Bond Energy (kJ/mol)
Br – Br	193
Cl – Cl	243
Br – Cl	?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

- ☐ · The response indicates I_2 .
- ☐ · The response gives an explanation such as the following.
 I_2 has the longest bond length because the radius of the I atom is greater than the radii of the other halogen atoms. Thus, the distance between the nuclei of atoms in I_2 is greater than it is in smaller halogens.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria.

- ☐ The response includes the equation $2 \text{ Br}^- + \text{Cl}_2 \rightarrow \text{Br}_2 + 2 \text{ Cl}^-$.
- ☐ The response includes a correct calculation of cell potential, similar to the following.
 $E^\circ = E^\circ(\text{reduced species}) - E^\circ(\text{oxidized species})$
 $= 1.36 \text{ V} - 1.07 \text{ V} = +0.29 \text{ V}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 1: Galvanic Cell



0	1	2
---	---	---

The response meets both of the following criteria.

- ☐ The response indicates that the only intermolecular attractions in $\text{Br}_2(l)$ are London forces, while those in $\text{BrCl}(l)$ include both London forces and dipole-dipole forces.
- ☐ The response indicates that due to the greater polarizability of the electron cloud of Br_2 compared to that of BrCl , the London forces in $\text{Br}_2(l)$ are stronger than the combined intermolecular forces in $\text{BrCl}(l)$ and thus the boiling point of $\text{Br}_2(l)$ is greater than that of $\text{BrCl}(l)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response includes the following calculation.

$$P = \frac{nRT}{V} = \frac{(0.100 \text{ mol})(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(298 \text{ K})}{2.00 \text{ L}} = 1.22 \text{ atm}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets one of the following criteria.

- ☐ The response gives the expression $K_{eq} = \frac{[\text{Br}_2][\text{Cl}_2]}{[\text{BrCl}]^2}$.
- ☐ The response gives the expression $K_{eq} = \frac{P_{\text{Br}_2}P_{\text{Cl}_2}}{(P_{\text{BrCl}})^2}$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. Solutions in terms of molar concentrations can also earn full credit. The response should be consistent with the expression in part (e).

Chapter 12 Part 1: Galvanic Cell



0	1	2
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The response meets both of the following criteria.

- ☐ The response meets both of the following criteria.
 $P_{\text{BrCl decomposed}} = (0.42)(1.22 \text{ atm}) = 0.51 \text{ atm}$
 $P_{\text{BrCl remaining}} = 1.22 \text{ atm} - 0.51 \text{ atm} = 0.71 \text{ atm}$
 $2x = 0.51 \text{ atm} \Rightarrow x = 0.26 \text{ atm Br}_2(g) \text{ and } 0.26 \text{ atm Cl}_2(g)$
- ☐ The response shows the following calculation.
 $K_{eq} = \frac{(0.26)(0.26)}{(0.71)^2} = 0.13$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows a correct calculation of the bond energy similar to the following.

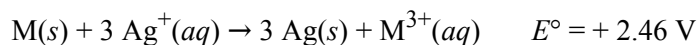
$$\Delta H^\circ = \Sigma(\text{bond energies})_{\text{broken}} - \Sigma(\text{bond energies})_{\text{formed}}$$

$$1.6 \text{ kJ/mol} = 2(\text{Br} - \text{Cl bond energy}) - (193 \text{ kJ/mol} + 243 \text{ kJ/mol})$$

$$(436 + 1.6) \text{ kJ/mol} = 2(\text{Br} - \text{Cl bond energy})$$

$$\text{Br} - \text{Cl bond energy} = 219 \text{ kJ/mol}$$

32.



According to the information above, what is the standard reduction potential for the half-reaction $\text{M}^{3+}(aq) + 3 e^- \rightarrow \text{M}(s)$?

Chapter 12 Part 1: Galvanic Cell

(A) -1.66 V



(B) -0.06 V

(C) 0.06 V

(D) 1.66 V

(E) 3.26 V

33.

Half-Reaction	E° (V)
$\text{Mg}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Mg}(\text{s})$	-2.37
$\text{Cr}^{3+}(\text{aq}) + 3 e^- \rightarrow \text{Cr}(\text{s})$	-0.74

Based on the information in the table above, which of the following shows the cell potential and the Gibbs free energy change for the overall reaction that occurs in a standard galvanic cell?

(A)	E°_{cell} (V)	ΔG° (kJ / mol _{rxn})
	+1.63	-157

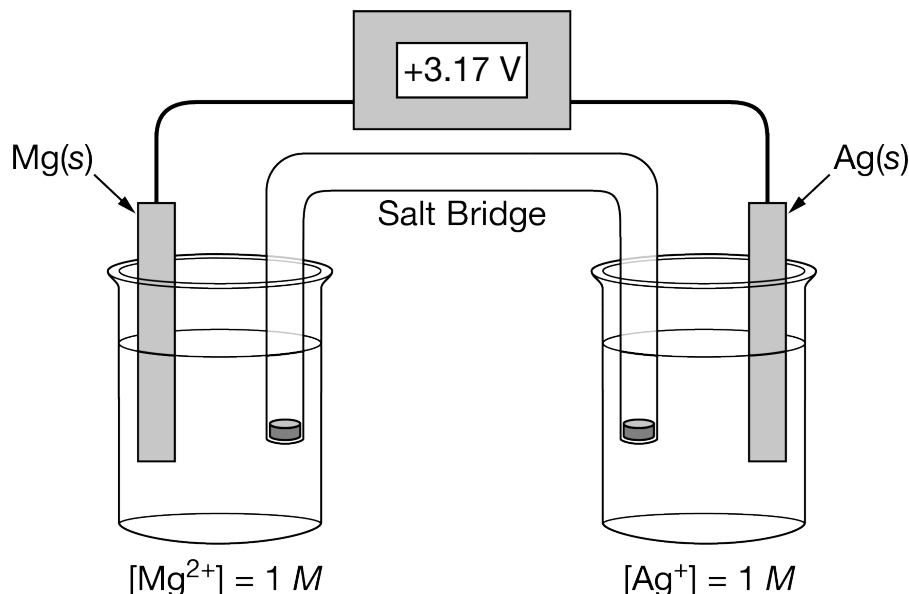
(B)	E°_{cell} (V)	ΔG° (kJ / mol _{rxn})
	+1.63	-944

(C)	E°_{cell} (V)	ΔG° (kJ / mol _{rxn})
	+5.63	-543

(D)	E°_{cell} (V)	ΔG° (kJ / mol _{rxn})
	+5.63	-3262

Chapter 12 Part 1: Galvanic Cell

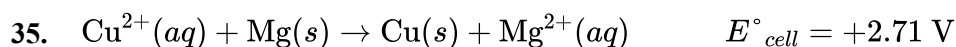
34.



Half-Reaction	Standard Reduction Potential, E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Mg}^{2+}(aq) + 2e^- \rightarrow \text{Mg}(s)$	−2.37

The galvanic cell shown above generates a cell potential of +3.17 V when operated under standard conditions. A second galvanic cell is made from the same two metals, and the measured cell potential is +3.25 V. Which of the following could be the reason for the second cell having a greater cell potential?

- (A) The mass of the Ag electrode in the second cell is greater than in first, resulting in a reduction potential greater than 0.80 V.
- (B) The mass of the Mg electrode in the second cell is smaller than in the first, resulting in a reduction potential lower than −2.37 V.
- (C) $[\text{Mg}^{2+}] > 1\text{ M}$ and $[\text{Ag}^+] = 1\text{ M}$ in the second cell, resulting in $Q > 1$, and $Q = 1$ for the first cell.
- (D) $[\text{Ag}^+] > 1\text{ M}$ and $[\text{Mg}^{2+}] = 1\text{ M}$ in the second cell, resulting in $Q < 1$, and $Q = 1$ for the first cell. ✓



A galvanic cell generates a potential of +2.71 V when $[\text{Cu}^{2+}] = [\text{Mg}^{2+}] = 1\text{ M}$ based on the chemical reaction represented above. Which of the following provides the best justification that the given conditions can be used to decrease the cell potential ($E_{\text{cell}} < +2.71\text{ V}$)?

Chapter 12 Part 1: Galvanic Cell

- (A) $[\text{Cu}^{2+}] = [\text{Mg}^{2+}] = 0.10\text{ M}$, because equilibrium is approached at a faster rate, causing a decrease in E_{cell} .
- (B) $[\text{Cu}^{2+}] = [\text{Mg}^{2+}] = 2.0\text{ M}$, because equilibrium is approached at a slower rate, causing a decrease in E_{cell} .
- (C) $[\text{Cu}^{2+}] = 1.0\text{ M}$ and $[\text{Mg}^{2+}] = 0.10\text{ M}$, because Q is smaller than 1, making E_{cell} decrease.
- (D) $[\text{Cu}^{2+}] = 0.10\text{ M}$ and $[\text{Mg}^{2+}] = 1.0\text{ M}$, because Q is greater than 1, making E_{cell} decrease. ✓

36.

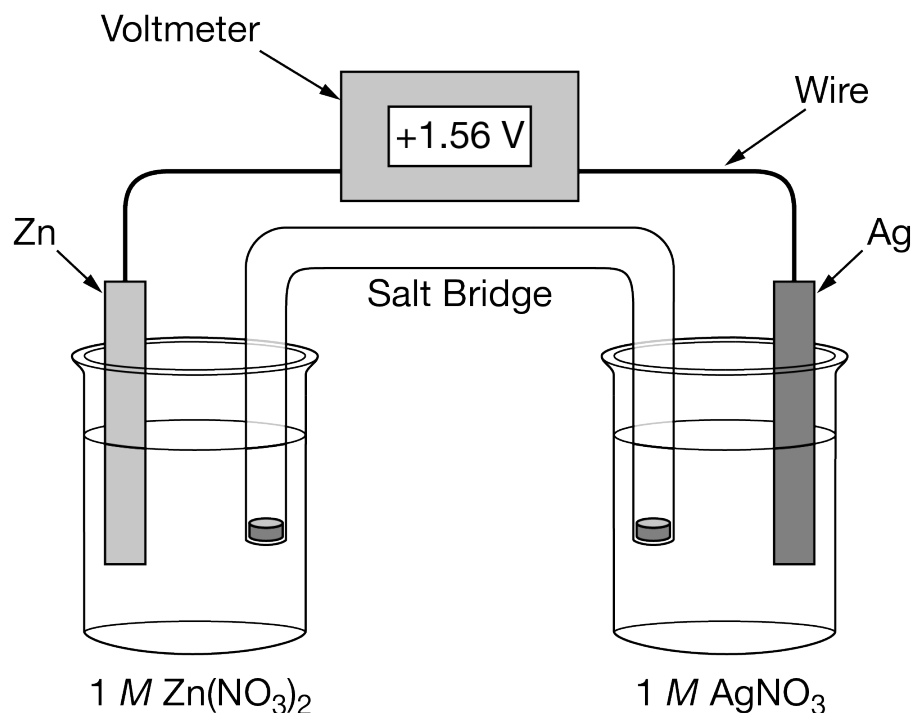
Reduction Half-Reaction	E° (V)
$\text{Ag}^+ + e^- \rightarrow \text{Ag}$	0.80
$\text{Sn}^{2+} + 2 e^- \rightarrow \text{Sn}$	-0.14
$\text{Fe}^{2+} + 2 e^- \rightarrow \text{Fe}$	-0.44
$\text{Zn}^{2+} + 2 e^- \rightarrow \text{Zn}$	-0.76

The standard reduction potentials of some half-reactions are given in the table above. Of the following equations, which represents the reaction that would be the basis of a cell with the most positive E_{cell}° ?

- (A) $2\text{Ag}^+ + \text{Sn} \rightarrow 2\text{Ag} + \text{Sn}^{2+}$ ✓
- (B) $\text{Sn}^{2+} + 2\text{Ag} \rightarrow \text{Sn} + 2\text{Ag}^+$
- (C) $\text{Fe}^{2+} + \text{Zn} \rightarrow \text{Fe} + \text{Zn}^{2+}$
- (D) $\text{Zn}^{2+} + \text{Fe} \rightarrow \text{Zn} + \text{Fe}^{2+}$

Chapter 12 Part 1: Galvanic Cell

37.



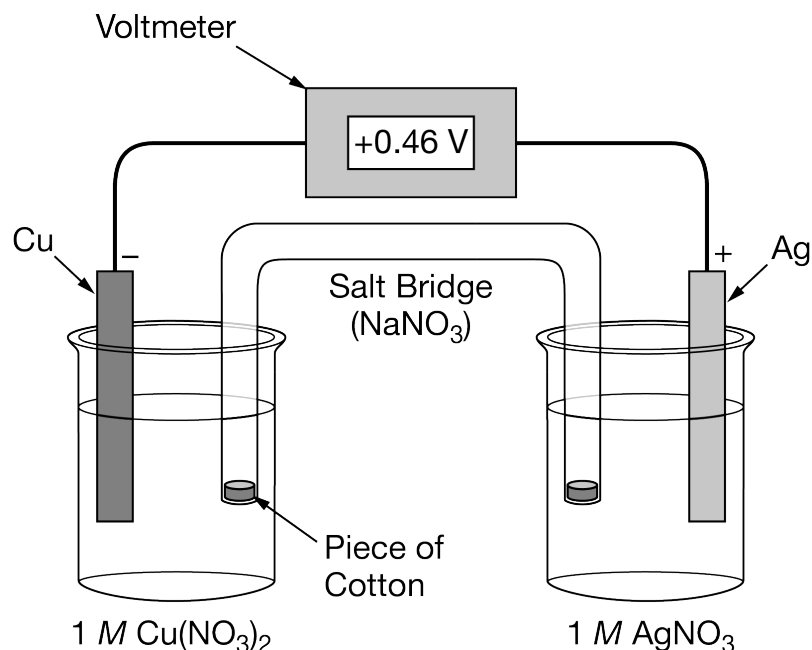
Half-Reaction	Standard Reduction Potential, E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Pb}^{2+}(aq) + 2e^- \rightarrow \text{Pb}(s)$	-0.13
$\text{Zn}^{2+}(aq) + 2e^- \rightarrow \text{Zn}(s)$	-0.76

The cell potential for the standard galvanic cell shown above is +1.56 V. If $\text{AgNO}_3(aq)|\text{Ag}(s)$ is replaced with $1\text{ M Pb}(\text{NO}_3)_2(aq)$ solution and a Pb electrode, which of the following describes what happens to the operation of the cell, and why?

- (A) Nothing changes because galvanic cells that have a $\text{Zn}(s)$ electrode have a constant cell potential, E°_{cell} , of +1.56 V.
- (B) The cell stops generating a voltage because the standard reduction potentials of Pb^{2+} and Zn^{2+} are both negative.
- (C) The cell potential decreases because the reduction of Pb^{2+} is less thermodynamically favorable than the reduction of Ag^+ . ✓
- (D) The cell potential increases because twice as many electrons are transferred between Pb^{2+} and Zn than between Ag^+ and Zn.

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38.



To construct the galvanic cell illustrated above, the salt bridge was prepared by soaking a piece of cotton in $5.0\text{ M NaNO}_3(aq)$ before placing it inside the U-shaped tube filled with distilled water. If the cotton was soaked in distilled water by mistake, which of the following best explains how the operation of the cell would be affected?

- (A) The operation of the cell is not affected because neither $\text{Na}^+(aq)$ nor $\text{NO}_3^-(aq)$ is involved in the redox reaction that takes place.
- (B) The operation of the cell generates a higher potential because there are fewer ions in the solution, making the reaction more thermodynamically favored.
- (C) The cell will operate for a much longer time because the flow of electrons through the circuit will eventually be reversed.
- (D) The cell would not operate because a current could not be conducted between the half-cells. ✓

39.

Half-Reaction	E° (V)
$\text{Mn}^{2+}(aq) + 2e^- \rightarrow \text{Mn}(s)$	-1.18
$\text{Sc}^{3+}(aq) + 3e^- \rightarrow \text{Sc}(s)$	-2.08

Based on the reduction potentials given in the table above, which of the following gives the balanced chemical equation and the correct standard cell potential for a galvanic cell involving $\text{Sc}^{3+}(aq)$ and $\text{Mn}^{2+}(aq)$?

- (A) $2\text{Sc}^{3+}(aq) + 3\text{Mn}(s) \rightleftharpoons 2\text{Sc}(s) + 3\text{Mn}^{2+}(aq)$ $E^\circ = -0.90\text{ V}$
- (B) $2\text{Sc}^{3+}(aq) + 3\text{Mn}(s) \rightleftharpoons 2\text{Sc}(s) + 3\text{Mn}^{2+}(aq)$ $E^\circ = -0.62\text{ V}$
- (C) $2\text{Sc}(s) + 3\text{Mn}^{2+}(aq) \rightleftharpoons 2\text{Sc}^{3+}(aq) + 3\text{Mn}(s)$ $E^\circ = +0.62\text{ V}$
- (D) $2\text{Sc}(s) + 3\text{Mn}^{2+}(aq) \rightleftharpoons 2\text{Sc}^{3+}(aq) + 3\text{Mn}(s)$ $E^\circ = +0.90\text{ V}$ ✓

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40.

Half-Reaction	E° (V)
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	0.80
$\text{Cr}^{3+}(aq) + 3e^- \rightarrow \text{Cr}(s)$	-0.41

Based on the standard reduction potentials in the table above, what is the value of E° for a standard galvanic cell made with Ag/Ag^+ and Cr/Cr^{3+} half-cells?

(A) 0.39 V

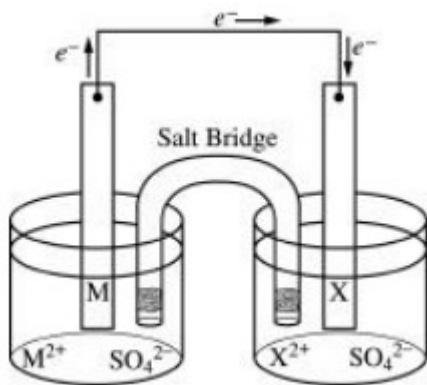
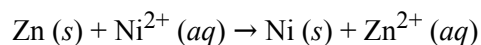
(B) 1.21 V

(C) 1.99 V

(D) 2.81 V



41. The diagram below shows the experimental setup for a typical electrochemical cell that contains two standard half-cells. The cell operates according to the reaction represented by the following equation.



- Identify M and M^{2+} in the diagram and specify the initial concentration for M^{2+} in solution.
- Indicate which of the metal electrodes is the cathode. Write the balanced equation for the reaction that occurs in the half-cell containing the cathode.
- What would be the effect on the cell voltage if the concentration of Zn^{2+} was reduced to 0.100 M in the half-cell containing the Zn electrode?
- Describe what would happen to the cell voltage if the salt bridge was removed. Explain.

Part A

1 point earned for correct M and M^{2+}

Chapter 12 Part 1: Galvanic Cell

1 point for the correct concentration of M^{2+} (Zn^{2+})

Electrons flow from the anode to the cathode in a voltaic electrochemical cell. The anode is where oxidation occurs, and in the reaction above, $Zn(s)$ is oxidized. So, the anode electrode must be $Zn(M)$ and the solution contains Zn^{2+} (M^{2+}). The $[Zn^{2+}] = 1.0\text{ M}$ in a standard cell. Additionally, the reduction potential for the Zn^{2+}/Zn redox couple is less than that for Ni^{2+}/Ni .



0	1	2
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The student response earns two of the following points:

1 1 point earned for correct M and M^{2+}

1 point for the correct concentration of M^{2+} (Zn^{2+})

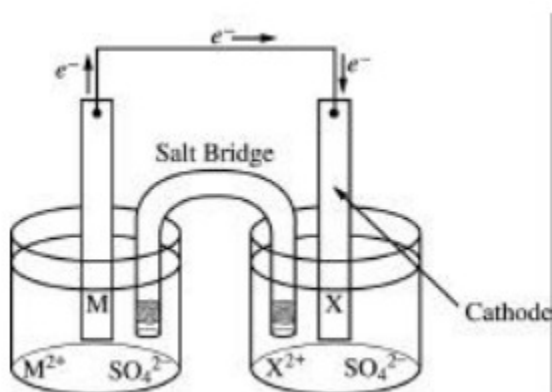
Electrons flow from the anode to the cathode in a voltaic electrochemical cell. The anode is where oxidation occurs, and in the reaction above, $Zn(s)$ is oxidized. So, the anode electrode must be $Zn(M)$ and the solution contains Zn^{2+} (M^{2+}). The $[Zn^{2+}] = 1.0\text{ M}$ in a standard cell. Additionally, the reduction potential for the Zn^{2+}/Zn redox couple is less than that for Ni^{2+}/Ni .

Part B

1 point earned for labeling the cathode in the cell diagram

1 point earned for the half-reaction

The cathode is $Ni(s)$, indicated by “X”



The half-reaction is $2e^{-} + Ni^{2+} \rightarrow Ni(s)$.



0	1	2
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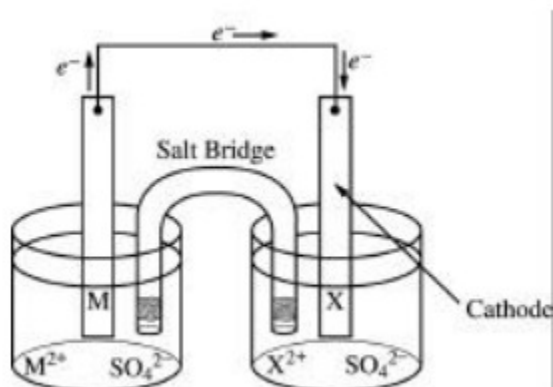
Chapter 12 Part 1: Galvanic Cell

The student response earns two of the following points:

1 point earned for labeling the cathode in the cell diagram

1 point earned for the half-reaction

The cathode is Ni(s), indicated by “X”



The half-reaction is $2e^- + \text{Ni}^{2+} \rightarrow \text{Ni(s)}$.

Part C

1 point earned for indicating that E_{cell} increases (is larger)

1 point earned for recognizing that $Q < 1$ and/or that the term

$$\frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Ni}^{2+}]}$$

must be added to the E°

$$E_{\text{cell}} = E^\circ -$$

$$\frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Ni}^{2+}]}$$

When the $[\text{Zn}^{2+}]$ is lowered to 0.100 M, then $Q < 1$. The value of the cell potential under these nonstandard conditions is more positive than E° (under standard conditions). The cell voltage increases.

An argument involving LeChâtelier's principle is also acceptable: the decreased $[\text{Zn}^{2+}]$ increases the “potential” for the reaction to proceed to the right.



0	1	2
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The student response earns two of the following points:

Chapter 12 Part 1: Galvanic Cell

1 point earned for indicating that E_{cell} increases (is larger)

1 point earned for recognizing that $Q < 1$ and/or that the term

$$\frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Ni}^{2+}]}$$

must be added to the E°

$$E_{\text{cell}} = E^\circ -$$

$$\frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Ni}^{2+}]}$$

When the $[\text{Zn}^{2+}]$ is lowered to 0.100 M, then $Q < 1$. The value of the cell potential under these nonstandard conditions is more positive than E° (under standard conditions). The cell voltage increases.

An argument involving LeChâtelier's principle is also acceptable: the decreased $[\text{Zn}^{2+}]$ increases the “potential” for the reaction to proceed to the right.

Part D

1 point earned for the effect

1 point earned for the explanation

The cell voltage drops to zero when the salt bridge is removed.

This happens because the salt bridge is needed to allow charge balance to occur in the solutions the electrodes are immersed in. In the absence of the salt bridge, ions cannot flow to balance the buildup of cations in the anode compartment and the buildup of anions in the cathode compartment.



0	1	2
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The student response earns two of the following points:

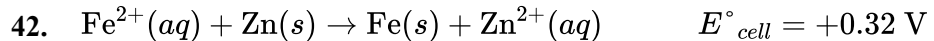
1 point earned for the effect

1 point earned for the explanation

The cell voltage drops to zero when the salt bridge is removed.

This happens because the salt bridge is needed to allow charge balance to occur in the solutions the electrodes are immersed in. In the absence of the salt bridge, ions cannot flow to balance the buildup of cations in the anode compartment and the buildup of anions in the cathode compartment.

Chapter 12 Part 1: Galvanic Cell



A galvanic cell generates a cell potential of $+0.32 \text{ V}$ when operated under standard conditions according to the reaction above. Which of the following pairs of conditions are needed to construct a similar cell that generates the lowest cell potential?

(A) $[\text{Zn}^{2+}] = 0.5 \text{ M}$ and $[\text{Fe}^{2+}] = 1 \text{ M}$ because $Q < 1$.

(B) $[\text{Zn}^{2+}] = 1 \text{ M}$ and $[\text{Fe}^{2+}] = 2 \text{ M}$ because $Q > 1$.

(C) $[\text{Zn}^{2+}] = 2 \text{ M}$ and $[\text{Fe}^{2+}] = 0.5 \text{ M}$ because $Q > 1$. ✓

(D) $[\text{Zn}^{2+}] = 2 \text{ M}$ and $[\text{Fe}^{2+}] = 1 \text{ M}$ because $Q < 1$.